



Key Features

Processor

- XuanTie C906 RISC-V CPU

Memory

- SiP 64 MB DDR2
- SD3.0, eMMC4.41

Video Engine

- Video decoding
 - H.265 up to 1920x1080@60fps
 - H.264 up to 1920x1080@60fps
 - H.263, MPEG-1/2/4, VC-1, up to 1080p@60fps
- Image decoding
 - JPEG up to 1080p@60fps
- Video encoding
 - JPEG/MJPEG up to 1920x1080@60fps
 - Maximum resolution: 8192x8192
 - Supports input picture scaling. The scaling ratio for width and height is 0.25 to 8.

Video Input

- 8-bit parallel CSI interface

Video Output

- 1 x RGB888 output interface up to 1920 x 1080@60fps
- Dual link LVDS interface up to 1920 x 1080@60fps
- 4-lane MIPI DSI interface up to 1920 x 1200@60fps (reduced blanking)

Audio

- 2 DACs and 2 ADCs
- Analog audio interfaces: MICIN1P/N, FMINL/R, LINEINL/R, LINEOUTLP/N, LINEOUTRP/N, HPOUTL/R
- Digital audio interfaces: I2S/PCM, DMIC, OWA IN/OUT

Peripherals Interfaces

- USB2.0 DRD x 1
- USB2.0 Host x 1
- SDIO 3.0, LEDC, IR TX&RX
- SPI x 2
- UART x 6
- TWI x 4
- PWM (8-ch)
- GPADC (2-ch)
- TPADC (4-ch, TPADC pad can be used by GPIO)
- 2 x 10/100/1000M Ethernet port with RMII and RGMII interfaces

Security Subsystem

- AES, DES, and 3DES encryption and decryption algorithms
- MD5, SHA, and HMAC tamper proofing
- RSA signature and verification algorithms

Device Summary

Orderable Device(s)	Package
F136M2-XXX	QFP 128 pins Body size: 14 mm x 14 mm Maximum height: 1.60 mm

Revision History

Revision	Date	Author	Description
0.90	September 13, 2025	AWA1896	Initial Version

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About This Document

Purpose and Scope

The documentation describes features of each module, pin/signal characteristics, current consumption, interface timing, thermal and package of the F136 family. For details about register descriptions of each module, see the *F136_User_Manual*.

Intended Audience

The document is intended for:

- Hardware designers and maintenance personnel for electronics
- Sales personnel for electronic parts and components

Related Documentation

For a complete listing of related documentation and development-support tools for the device, contact the Allwinner FAE or access the Allwinner Customer Service Platform by visiting <https://open.allwinnertech.com/>.

Revision Number Definition

Revision 0.90-0.9x

This document is released based on the design completion products yet to be mass-produced. Therefore, the information in this document may be modified by reason of mass-produced verification.

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If you have any questions about the document, please contact us to confirm and obtain the latest version.

Conventions

Symbol Conventions

The symbols that may be found in this document are defined as follows.

Symbol	Description
 CAUTION	A caution means that damage to equipment is possible.
 NOTE	Provides additional information to emphasize or supplement important points of the main text.

Table Content Conventions

The table content conventions that may be found in this document are defined as follows.

Symbol	Description
-	The cell is blank.

Numerical System

The expressions of the data capacity, the frequency, and the data rate are described as follows.

Type	Symbol	Value
Data capacity	K	1024
	M	1,048,576
	G	1,073,741,824
Frequency, data rate	k	1000
	M	1,000,000
	G	1,000,000,000

FT/QA/QC Test

All Allwinner chips provided for clients have passed the following tests.

Test Item	Description
FT Test	FT test is the finished product testing after the chip is packaged, and it is a functional test of all modules for each produced chip.
QA Test	QA test is a system-level sampling test for good-quality chips. According to the application level of the chip, a certain percentage of good-quality chips are selected for system-level testing to make the chip work in a typical application scenario, and judge whether the chip works normally in this scenario.

Test Item	Description
QC Test	QC test is a module-level sampling test for good-quality chips. According to the chip application level, a certain percentage of good-quality chips are selected for module-level functional testing to monitor whether the chip production process is normal.

1 Overview

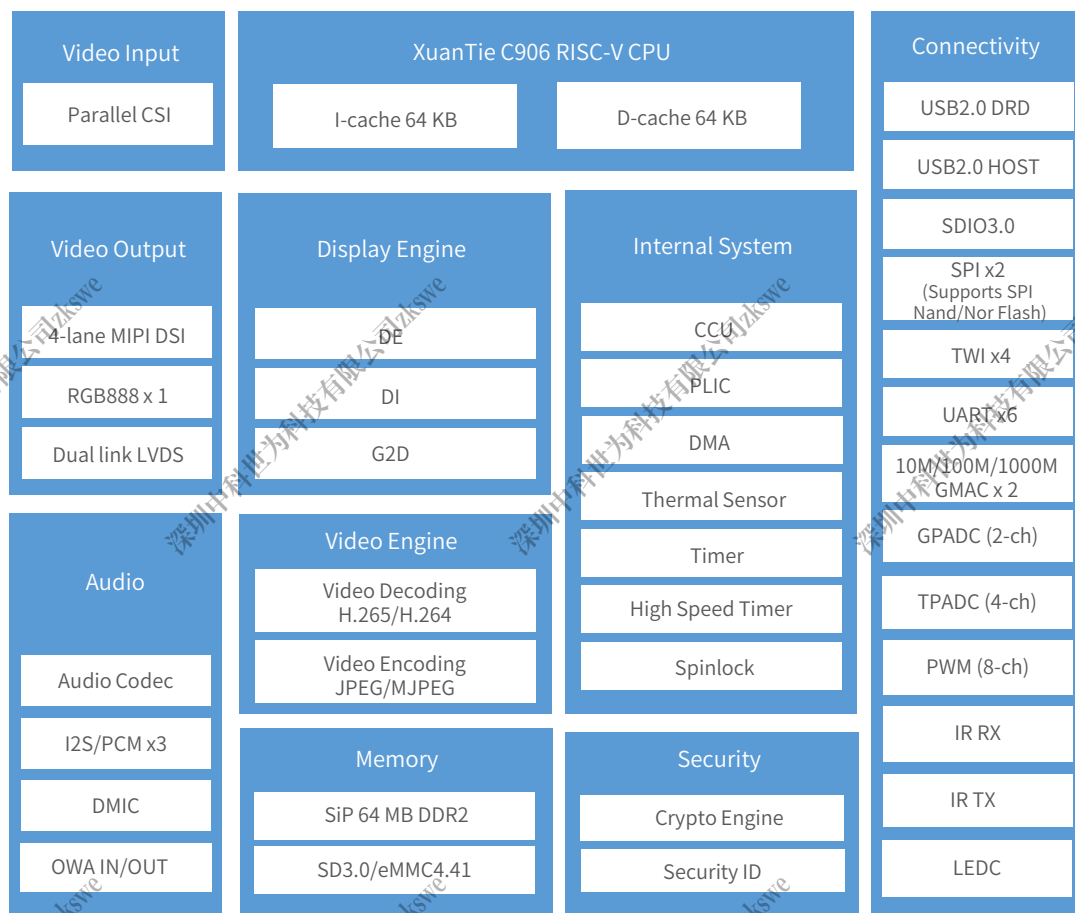
The F136 family is an advanced application processor designed specifically for the video decoding market. It has a rich set of peripheral interfaces and many new features to provide a low-cost overall solution design.

The F136 has the following highlights:

- Integrated 64-bit RISC CPU processor provides powerful computing performance.
- Embedded 64 MB DDR2, up to 528MHz.
- Supports H.265/H.264 1080p@60fps video decoding, and MJPEG/JPEG 1080p@60fps video encoding.
- Rich peripheral interfaces, such as USB, SDIO, GMAC, UART, SPI, PWM, GPADC, TPADC, IR, and so on.
- Rich video interfaces, RGB888, Dual-LVDS, MIPI-DSI for Display, DVP-CSI for video input.

The following figure shows the system block diagram for the device.

Figure 1-1 System Block Diagram



2 Features

2.1 Processors

- XuanTie C906 RISC-V CPU
- 64 KB I-cache + 64 KB D-cache

2.2 Memory

2.2.1 Boot ROM (BROM)

- On-chip memory
- Supports system boot from the following devices:
 - SD Card
 - eMMC
 - SPI NOR Flash (Quad Mode and Single Mode)
 - SPI NAND Flash
- Supports mandatory upgrade process through USB or SD card
- Supports GPIO pin and eFuse module to select the boot media type
- Supports BROM loads only certified firmware

2.2.2 SDRAM

- SiP 64 MB DDR2

2.2.3 SD/MMC Host Controller (SMHC)

- Three SD/MMC Host Controller (SMHC) interfaces
 - SMHC0, compliant with the protocol Secure Digital Memory (up to SD3.0)
 - SMHC1, compliant with the protocol Secure Digital I/O (up to SDIO3.0)
 - SMHC2, compliant with the protocol Multimedia Card (up to eMMC4.41)
- The SMHC0 and the SMHC1 support the following:
 - 1-bit or 4-bit data width
 - Maximum performance:
 - SDR mode 150 MHz@1.8 V IO pad
 - DDR mode 50 MHz@1.8 V IO pad
 - SDR mode 50 MHz@3.3 V IO pad
- The SMHC2 supports the following:
 - 1-bit or 4-bit data width
 - Maximum performance:

- SDR mode 50 MHz@3.3V IO pad
- DDR mode 50 MHz@3.3V IO pad
- Supports block size of 1 to 65535 bytes
- Supports hardware CRC generation and error detection
- Internal 1024-Bytes RX FIFO and 1024-Bytes TX FIFO
- Supports descriptor-based internal DMA controller

2.3 Graphics Processing

2.3.1 Display Engine (DE)

- Output size up to 2048 x 2048
- Supports two alpha blending channels for main display
- Supports four overlay layers in each channel, and has an independent scaler
- Supports potter-duff compatible blending operation
- Support LBC buffer decoder for YUV420
- Supports dither output to TCON
- Supports input format Semi-planar YUV422/YUV420/YUV411 and Planar YUV422/YUV420/YUV411, ARGB8888/XRGB8888/RGB888/ARGB4444/ARGB1555/RGB565/palette
- Supports Awonder1.1 Lite for excellent display experience
 - Adaptive luminance de-noising for compression noise or mosquito noise for yuv420/422 input
 - Adaptive horizontal detail/edge enhancement
 - Hue gain, Saturation gain, and Value gain controlled
 - Fully programmable color matrix
 - Dynamic gamma
 - 17x17x17 programmable LUT for 3D color management
 - global dynamic contrast enhancement
 - Color temperature tuning/Auto white balance
- Supports write back only for verification
- Support RCQ to configure registers
- Support keystone correction

2.3.2 De-interlacer (DI)

- Supports YUV420 (Planar/NV12/NV21) and YUV422 (Planar/NV16/NV61) data format
- Supports video resolution from 32 x 32 to 2048 x 1280 pixel
- Supports Inter-field interpolation/motion adaptive de-interlace method
- Performance: module clock 600 MHz for 1080p@60Hz YUV420

2.3.3 Graphic 2D (G2D)

- Supports mixer layer size up to 2048 x 2048 pixels, and rotation size up to 4096x4096 pixels
- Supports pre-multiply alpha image data
- Supports color key
- Supports two pipes Porter-Duff alpha blending
- Supports multiple video formats 4:2:0, 4:2:2, 4:1:1 and multiple pixel formats (8/16/24/32 bits graphics layer)
- Supports memory scan order option
- Supports any format convert function
- Supports 1/16× to 32× resize ratio
- Supports 32-phase 8-tap horizontal anti-alias filter and 32-phase 4-tap vertical anti-alias filter
- Supports window clip
- Supports FillRectangle, BitBlit, StretchBlit and MaskBlit
- Supports horizontal and vertical flip, clockwise 0/90/180/270 degree rotate for normal buffer

2.4 Video Engine

2.4.1 Video Decoder

2.4.1.1 Video Decoding

- H.265/HEVC Main Profile@Level4.1, up to 1920 x 1080@60fps
- H.264/AVC, BP/MP/HP@Level4.2, up to 1920 x 1080@60fps
- H.263 BP, up to 1920 x 1080@60fps
- MPEG-4 SP@L0-3, ASP@L0-5, up to 1920 x 1080@60fps
- MPEG-2 SP@ML, MP@HL, up to 1920 x 1080@60fps
- MPEG-1, up to 1920 x 1080@60fps
- VC-1 SP@ML, MP@HL, AP@L0-4, up to 1920 x 1080@60fps

2.4.1.2 Image Decoding

- JPEG decoding
 - Up to 1920 x 1080@60fps
 - Maximum decoding resolution: 1920x1088

2.4.2 Video Encoder

- JPEG/MJPEG up, to 1920x1080@60fps
- Maximum resolution: 8192x8192
- Supports input picture scaling. The scaling ratio for width and height is 0.25 to 8

2.5 Video Input Interfaces

2.5.1 Parallel CSI

- Supports 8-bit digital camera interface (RAW8/YUV422/YUV420)
- Supports BT656, BT601 interface (YUV422)
- Supports ITU-R BT.656 time-multiplexed format up to 2*1080p@30fps in DDR sample mode
- Maximum pixel clock of 148.5 MHz
- Supports de-interlacing for interlace video input
- Supports horizontal and vertical flip

2.6 Video Output Interfaces

2.6.1 LCD

- One panel interface
 - RGB interface with DE/SYNC mode, up to 1920 x 1080@60fps
 - Serial/dummy RGB interface, up to 800 x 480@60fps
 - Dither function for RGB888, RGB666 and RGB565
 - Support BT.656 interface for NTSC and PAL
 - i8080 interface, up to 800x480@60fps
 - Dual-link LVDS interface
 - Up to 1920x1080@60fps for dual-link
 - Up to 1366x768@60fps for single-link

2.6.2 MIPI DSI

- Compliant with MIPI DSI specification v1.01 and MIPI D-PHY specification v1.0
- Up to 1.0 Gbit/s per lane
- Supports one 4-lane MIPI DSI display, up to 1920 x 1200@60fps (reduced blanking)

2.7 System Peripherals

2.7.1 Clock Controller Unit (CCU)

- Supports clock configuration and clock generation for corresponding modules
- Supports software-controlled clock gating and software-controlled reset for corresponding modules

2.7.2 Power Reset Clock Management (PRCM)

- Module clock configuration
- Bus gating, bus reset, and clock configuration

- RAM configuration control

2.7.3 RTC

- One on-chip RC oscillator
- Supports one external 24 MHz DCXO
- Provides a 16-bit counter for counting day, 5-bit counter for counting hour, 6-bit counter for counting minute, 6-bit counter for counting second
- Timer frequency is 1 kHz
- Configurable initial value by software anytime
- Supports timing alarm, and generates interrupt and wakeup the external devices
- 8 general purpose registers for storing power-off information in RTC domain

2.7.4 Thermal Sensor Controller (THS)

- One thermal sensor
- Averaging filter for thermal sensor reading
- Supports over-temperature protection interrupt and over-temperature alarm interrupt
- Support $\pm 3^{\circ}\text{C}$ error limit from 60°C to 125°C , and $\pm 5^{\circ}\text{C}$ from -40°C to 60°C

2.7.5 LDO Power

- Integrated 1 LDO(LDOA)
- LDOA: 1.8 V power output
- LDOA for IO and analog module
- Input voltage is 2.8 V to 3.6 V

2.7.6 Timer

- Supports Timer0 and Timer1
- Timer0 supports the following features:
 - The timer module implements the timing and counting functions, which includes timer0, timer1, watchdog, and audio video synchronization (AVS)
 - The timer0/timer1 is a 32-bit down counter. The timer0 and timer1 are completely consistent
 - The watchdog is used to transmit a reset signal to reset the entire system when an exception occurs in the system
 - The AVS is used to synchronize the audio and video. The AVS sub-block includes AVS0 and AVS1, which are completely consistent
- Timer1 contains timer0-3. The timer0/timer1/timer2/timer3 is a 32-bit down counter. They are completely consistent.

2.7.7 High Speed Timer (HSTimer)

- The HSTimer module consists of HSTimer0 and HSTimer1. HSTimer0 and HSTimer1 are down counters that implement timing and counting functions. They are completely consistent.
- Configurable 56-bit down timer
- Supports 5 prescale factors
- The clock source is synchronized with AHB0 clock, much more accurate than other timers
- Supports 2 working modes: periodic mode and single counting mode
- Generates an interrupt when the count is decreased to 0

2.8 Audio

2.8.1 Audio Codec

- Audio inputs
 - One differential microphone input: MICIN1P/N
 - One stereo LINEIN input: LINEINL/R
 - One stereo FMIN input: FMINL/FMINR
- Audio outputs
 - One stereo differential lineout output: LINEOUTLP/N, LINEOUTRP/N
 - One stereo headphone output: HPOUTL/R
- Two audio ADCs
 - 16-bit and 20-bit sample resolution
 - 8 kHz to 48 kHz ADC sample rate
 - 95 ± 3 dB SNR@A-weight, -80 ± 3 dB THD+N
- Two audio DACs
 - 16-bit and 20-bit sample resolution
 - 8 kHz to 192 kHz DAC sample rate
 - 100 ± 2 dB SNR@A-weight, -85 ± 3 dB THD+N
- Stereo headphone driver
 - 95 ± 3 dB SNR@A-weight
 - Output Level $0.54 \text{ Vrms}@10 \text{ k}\Omega/\text{THD+N}$ -77 ± 3 dB, $0.37 \text{ Vrms}@16 \text{ }\Omega/\text{THD+N}$ -40 dB
- Supports Dynamic Range Controller adjusting the DAC playback and ADC recording
- One 128x20-bits FIFO for DAC data transmit, one 128x20-bits FIFO for ADC data receive
- Programmable FIFO thresholds
- Supports interrupts and DMA

2.8.2 I2S/PCM

- Three I2S/PCM interfaces (I2S0, I2S1, and I2S2)
- Compliant with standard Philips Inter-IC sound (I2S) bus specification

- Left-justified, Right-justified, PCM mode, and TDM format
- Programmable PCM frame width: 1 BCLK width (short frame) and 2 BCLKs width (long frame)
- Transmit and receive data FIFOs
 - Programmable FIFO thresholds
 - 128 x 32-bit TXFIFO, 64 x 32-bit RXFIFO
- Supports multiple function clocks
 - Clock up to 24.576 MHz Data Output of I2S/PCM in Master mode (Only if the IO PAD and peripheral I2S/PCM satisfy timing parameters)
 - Clock up to 12.288 MHz Data Input of I2S/PCM in Master mode (Only if the IO PAD and peripheral I2S/PCM satisfy timing parameters)
- Supports TX/RX DMA slave interface
- Supports multiple application scenarios
 - Up to 16 channels ($f_s = 48 \text{ kHz}$) which has adjustable width from 8 bits to 32 bits
 - Sample rate from 8 kHz to 384 kHz (sample rate * channel * slot width $\leq 24.576 \text{ MHz}$)
 - 8-bit u-law and 8-bit A-law companded sample
- Supports master/slave mode

2.8.3 DMIC

- Supports maximum 8 digital PDM microphones
- Supports sample rate from 8 kHz to 48 kHz

2.8.4 One Wire Audio (OWA)

- One OWA TX and One OWA RX, compliant with S/PDIF interface
- Compatible with IEC-60958 and IEC-61937
 - IEC-60958 supports data formats: 16 bits, 20 bits, and 24 bits
- Transmit and receive data FIFOs
 - Programmable FIFO thresholds
 - One 128x24bits TXFIFO and one 64x24bits RXFIFO for audio data transfer
- Supports TX/RX DMA Slave interface
- Supports Multiple function clock
- Separate OWA TX and RX clock
 - OWA TX function clock includes a series of 24.576MHz and 22.579MHz frequency
 - OWA RX function clock includes a series of 24.576*8MHz frequency (RX function clock 24.576*8MHz support CDR of sample rate from 8KHz to 192KHz)
- Supports hardware parity On TX/RX
 - Hardware parity generation on the transmitter
 - Hardware parity checking on the receiver
- Supports channel status capture on the receiver

- Supports channel sample rate capture on the receiver
- Supports insertion detection for the receiver
- Support channel status insertion for the transmitter

2.9 Peripheral Interfaces

2.9.1 USB2.0 DRD

- One USB2.0 DRD (USB0), with integrated USB 2.0 analog PHY
- Complies with USB2.0 Specification
- Static host and device mode
- USB host that supports the following:
 - Compatible with Enhanced Host Controller Interface (EHCI) specification, version 1.0
 - Compatible with Open Host Controller Interface (OHCI) specification, version 1.0a
 - High-Speed (HS, 480 Mbit/s), Full-Speed (FS, 12 Mbit/s), and Low-Speed (LS, 1.5 Mbit/s)
 - Supports only 1 USB Root Port shared between EHCI and OHCI
- USB device that supports the following:
 - High-Speed (HS, 480 Mbit/s), Full-Speed (FS, 12 Mbit/s)
 - Up to 8KB+64Bytes FIFO for EPs (including EP0)
- Supports an internal DMA controller for data transfer with memory
- Device and Host controller share an 4KB SRAM and a physical PHY

2.9.2 USB2.0 Host

- One USB 2.0 Host (USB1), with integrated USB 2.0 analog PHY
- Compatible with Enhanced Host Controller Interface (EHCI) specification, version 1.0
- Compatible with Open Host Controller Interface (OHCI) specification, version 1.0a
- Supports High-Speed (HS, 480 Mbit/s), Full-Speed (FS, 12 Mbit/s) and Low-Speed (LS, 1.5 Mbit/s) Device
- An internal DMA Controller for data transfer with memory
- Supports only 1 USB Root Port shared between EHCI and OHCI

2.9.3 GMAC

- Two 10/100/1000 Mbit/s Ethernet port with RGMII and RMII interfaces (RGMII0/RMII0, RGMII1/RMII1)
- Compliant with IEEE 802.3-2002 standard
- Supports both full-duplex and half-duplex operations
- Provides the management data input/output (MDIO) interface for PHY device configuration and management with configurable clock frequencies

- Programmable frame length to support Standard or Jumbo Ethernet frames with sizes up to 16 KB
- Supports a variety of flexible address filtering modes
- Separate 32-bit status returned for transmission and reception packets
- Optimization for packet-oriented DMA transfers with frame delimiters
- Supports linked-list descriptor list structure
- Descriptor architecture, allowing large blocks of data transfer with minimum CPU intervention; each descriptor can transfer up to 4 KB of data
- Comprehensive status reporting for normal operation and transfers with errors
- 4 KB TXFIFO for transmission packets and 16 KB RXFIFO for reception packets
- Programmable interrupt options for different operational conditions

2.9.4 UART

- Up to 6 UART controllers (UART0, UART1, UART2, UART3, UART4, UART5)
- Compatible with industry-standard 16450/16550 UARTs
- 5-8 data bits, 1/1.5/2 stop bits, programmable parity (even, odd, or no parity)
- The working reference clock is from the APB bus clock
 - Speed up to 1.5 Mbit/s with 24 MHz APB clock
- Two separate FIFOs: one is RX FIFO, and the other is TX FIFO
 - 64 bytes RXFIFO and 64 bytes TXFIFO for UART0
 - 256 bytes RXFIFO and 256 bytes TXFIFO for UART1, UART2, UART3, UART4, UART5
- Supports 9-bit/RS-485 format and RS-485 full duplex mode
- Supports Software/hardware flow control
- Supports interrupts and DMA mode
- Supports auto-flow by using CTS & RTS for UART1, UART2, UART3

2.9.5 PWM

- Up to 8 PWM channels
 - Supports PWM continuous mode output
 - Supports PWM pulse mode output, and the pulse number is configurable
 - Output frequency range:
 - 0 to 24 MHz (when the clock source is DCXO)
 - 0 to 100 MHz (when the clock source is APB1 clock)
 - Various duty-cycle: 0% -100%
 - Minimum resolution: 1/65536
- Maximum 4 complementary pairs output
 - PWM01 pair (PWM-0 + PWM-1), PWM23 pair (PWM-2 + PWM-3), PWM45 pair (PWM-4 + PWM-5), PWM67 pair (PWM-6 + PWM-7)
 - Supports dead-zone generator, and the dead-zone time is configurable

- Maximum 4 group of PWM channel output for controlling stepping motors
 - Supports any plural channels to form a group, and output the same duty-cycle pulse
 - In group mode, the relative phase of the output waveform for each channel is configurable
- Maximum 8 channels capture input
 - Supports rising edge detection and falling edge detection for input waveform pulse
 - Supports pulse-width measurement for input waveform pulse

2.9.6 SPI and SPI_DBI

- Up to 2 SPI controllers
 - SPI0, supporting SPI mode
 - SPI1, supporting SPI mode and display bus interface (DBI) mode

SPI mode

- Multiple SPI modes:
 - Master mode and slave mode for standard SPI
 - Master mode for Dual-Output/Dual-Input SPI and Dual I/O SPI
 - Master mode for Quad-Output/Quad-Input SPI
 - Master mode for 3-wire SPI, with programmable serial data frame length of 1 bit to 32 bits
- Supports mode0, mode1, mode2, and mode3
- TX/RX DMA Slave interface
- Transmit data FIFO with 8-bit wide and 64-entry
- Receive data FIFO with 8-bit wide and 64-entry
- Transmit data buffer with 8-bit wide and 4-entry
- Receive data buffer with 8-bit wide and 128-entry
- Supports control signal configuration
 - One chip select
 - Polarity and phase of the Chip Select (SPI_CS) and SPI Clock (SPI_CLK) are configurable

DBI mode

- DBI Type C 3 Line/4 Line Interface Mode
- 2 Data Lane Interface Mode
- RGB111/444/565/666/888 video format
- Maximum resolution of RGB666 240 x 320@30Hz with single data lane
- Maximum resolution of RGB888 240 x 320@60Hz or 320 x 480@30Hz with dual data lane
- Tearing effect
- Software flexible control video frame rate

2.9.7 Two Wire Interface (TWI)

- Up to 4 TWI controllers (TWI0, TWI1, TWI2, TWI3)
- Compliant with I2C bus standard
- master mode and slave mode
- 7-bit and 10-bit device addressing modes
- Standard mode (up to 100 Kbit/s) and fast mode (up to 400 Kbit/s)
- Supports multi-master devices

2.9.8 General Purpose ADC (GPADC)

- One 2-ch ADC
- 12-bit sampling resolution
- Maximum sampling frequency up to 1 MHz
- Supports three operation modes: single conversion mode, continuous conversion mode, burst conversion mode
- Analog input range: 0 to 1.8 V

2.9.9 Touch Panel ADC (TPADC)

- 12-bit SAR type A/D converter
- 32x12 FIFO for storing A/D conversion result
- Supports DMA slave interface
- Supports AUX ADC, up to 4 channels
- 4-wire resistive touch panel input detection
 - Single touch detection
 - Dual touch detection
 - Touch pressure measurement with programmable threshold
 - Pen down detection with programmable sensitivity
 - Median and averaging filter for noise reduction
 - X/Y coordinate exchange
- Maximum sampling rate of 750 kHz
- Analog input range: 0 to 1.8 V

2.9.10 LED Controller (LEDC)

- LEDC is used to control the external intelligent control LED lamp
- Configurable LED output high/low level width and reset time
- LEDC data supports DMA configuration mode and CPU configuration mode
- Maximum 1024 LEDs serial connect
- Configurable interval time between data packets and frame data
- Configurable RGB display mode

- Configurable default level of non-data output

2.9.11 Consumer Infrared Transmitter (IR_TX)

- One IR_TX interface
- Supports arbitrary wave generator
- Configurable carrier frequency
- Supports handshake mode and waiting mode of DMA
- 128 bytes FIFO for data buffer

2.9.12 Consumer Infrared Receiver (IR_RX)

- One IR_RX interface
- NEC format infra data
- RLC encoding
- 64x8 bits FIFO for data buffer
- Sample clock up to 1 MHz

2.10 Security

2.10.1 Crypto Engine (CE)

- Supports Symmetrical algorithm for encryption and decryption: AES, DES, 3DES
 - Supports ECB, CBC, CTS, CTR, CFB, OFB mode for AES
 - Supports 128/192/256-bit key for AES
 - Supports ECB, CBC, CTR mode for DES/3DES
- Supports Hash algorithm for tamper proofing: MD5, SHA, HMAC
 - Supports SHA1, SHA224, SHA256, SHA384, SHA512 for SHA
 - Supports HMAC-SHA1, HMAC-SHA256 for HMAC
 - Supports multi-package mode for MD5/SHA1/SHA224/SHA256/SHA384/SHA512
- Supports Asymmetrical algorithm for signature verification: RSA
 - RSA supports 512/1024/2048-bit width
- Supports 160-bit hardware PRNG with 175-bit seed
- Supports 256-bit hardware TRNG
- Internal DMA controller for data transfer with memory

2.10.2 Security ID (SID)

- 2048-bit eFuse
- Backup eFuse information by using SID_SRAM
- One-time programming
- Selecting double-bit check by parameter definition

- Data scrambling
- Reading and writing protection

2.11 Package

QFP 128 pins, 14 mm x 14 mm body size, 1.60 mm height (maximum), 6.955 mm x 6.955 mm EPAD

3 Pin Description

3.1 Pin Characteristics

This section lists the characteristics of the device pins from the following seven aspects.

[1] **Pin#**: Package pin numbers associated with each signal.

[2] **Pin Name**: The name of the package pin.

NC means these pins are not connected.

[3] **Type**: Denotes the signal direction

I (Input),

O (Output),

I/O (Input/Output),

OD (Open-Drain),

A (Analog),

AI (Analog Input),

AO (Analog Output),

P (Power),

G (Ground)

N/A (Not Applicable)

[4] **Ball Reset State**: The state of the terminal at reset.

PU: Pull Up

PD: Pull Down

Z: High Impedance

N/A: Not Applicable

[5] **Pull Up/Down**: Denotes the presence of an internal pull-up or pull-down resistor. Pull-up and pull-down resistors can be enabled or disabled via software.

PU: Internal pullup

PD: Internal pulldown

PU/PD: Internal pullup and pulldown

N/A: Not Applicable

[6] **Default Buffer Strength**: Defines the default drive strength of the associated output buffer. The maximum drive strength of each GPIO is 6 mA.

N/A means Not Applicable.

[7] **I/O Power Supply**: The voltage supplies for the IO buffers of the terminal.

N/A means Not Applicable.

3.1.1 SDRAM

Table 3-1 SDRAM Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
64	VCC-DRAM0	P	N/A	N/A	N/A	N/A
67	VCC-DRAM1	P	N/A	N/A	N/A	N/A
66	VDD18-DRAM	P	N/A	N/A	N/A	N/A

3.1.2 System Control

Table 3-2 System Control Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
104	TEST	I	PD	PU/PD	N/A	VCC-RTC-PLL
105	RESET	I/O, OD	N/A	N/A	N/A	VCC-RTC-PLL

3.1.3 RTC&PLL

Table 3-3 RTC&PLL Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
106	NMI	I/O, OD	N/A	N/A	N/A	VCC-RTC-PLL
103	VCC-RTC-PLL	P	N/A	N/A	N/A	N/A

3.1.4 DCXO

Table 3-4 DCXO Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
107	DXIN	AI	N/A	N/A	N/A	VCC-RTC-PLL
108	DXOUT	AO	N/A	N/A	N/A	VCC-RTC-PLL
109	REFCLK	AO	N/A	N/A	N/A	VCC-RTC-PLL

3.1.5 GPIO Groups

3.1.5.1 Port A

Table 3-5 Port A Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
83	PA0	I/O	Z	PU/PD	4 mA	AVCC
82	PA1	I/O	Z	PU/PD	4 mA	AVCC
81	PA2	I/O	Z	PU/PD	4 mA	AVCC
80	PA3	I/O	Z	PU/PD	4 mA	AVCC

3.1.5.2 Port B

Table 3-6 Port B Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
27	PB1	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO
26	PB2	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO
25	PB3	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO
23	PB4	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO
22	PB5	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO
21	PB6	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO
20	PB7	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO

3.1.5.3 Port C

Table 3-7 Port C Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
78	PC0	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO
77	PC1	I/O	PU	PU/PD	4 mA	VCC33-USB-LDOIN-IO
76	PC2	I/O	PU	PU/PD	4 mA	VCC33-USB-LDOIN-IO
74	PC3	I/O	PU	PU/PD	4 mA	VCC33-USB-LDOIN-IO
75	PC4	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
73	PC5	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO
72	PC6	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO
71	PC7	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO
70	PC8	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO
69	PC9	I/O	Z	PU/PD	4 mA	VCC33-USB-LDOIN-IO

3.1.5.4 Port D

Table 3-8 Port D Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
4	PD0	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
5	PD1	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
6	PD2	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
7	PD3	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
8	PD4	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
9	PD5	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
12	PD6	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
13	PD7	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
14	PD8	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
15	PD9	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
122	PD10	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
123	PD11	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
124	PD12	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
125	PD13	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
126	PD14	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
127	PD15	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
128	PD16	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
1	PD17	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
2	PD18	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
3	PD19	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
16	PD20	I/O	Z	PU/PD	4 mA	VCC33-PD
17	PD21	I/O	Z	PU/PD	4 mA	VCC33-PD
18	PD22	I/O	Z	PU/PD	4 mA	VCC33-PD
10	VCC33-PD	P	N/A	N/A	N/A	N/A
11	VCC18-LVDS	P	N/A	N/A	N/A	N/A

3.1.5.5 Port E

Table 3-9 Port E Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
46	PE0	I/O	Z	PU/PD	4 mA	VCC-PE
47	PE1	I/O	Z	PU/PD	4 mA	VCC-PE
48	PE2	I/O	Z	PU/PD	4 mA	VCC-PE
50	PE3	I/O	Z	PU/PD	4 mA	VCC-PE
51	PE4	I/O	Z	PU/PD	4 mA	VCC-PE
52	PE5	I/O	Z	PU/PD	4 mA	VCC-PE
53	PE6	I/O	Z	PU/PD	4 mA	VCC-PE
54	PE7	I/O	Z	PU/PD	4 mA	VCC-PE
55	PE8	I/O	Z	PU/PD	4 mA	VCC-PE
56	PE9	I/O	Z	PU/PD	4 mA	VCC-PE
57	PE10	I/O	Z	PU/PD	4 mA	VCC-PE
58	PE11	I/O	Z	PU/PD	4 mA	VCC-PE
59	PE12	I/O	Z	PU/PD	4 mA	VCC-PE
60	PE13	I/O	Z	PU/PD	4 mA	VCC-PE
61	PE14	I/O	Z	PU/PD	4 mA	VCC-PE
62	PE15	I/O	Z	PU/PD	4 mA	VCC-PE
63	PE16	I/O	Z	PU/PD	4 mA	VCC-PE
49	VCC-PE	P	N/A	N/A	N/A	N/A

3.1.5.6 Port F

Table 3-10 Port F Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
115	PF0	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
116	PF1	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
117	PF2	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
118	PF3	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
119	PF4	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
120	PF5	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS
114	PF6	I/O	Z	PU/PD	4 mA	VCC33-PD/VCC18-LVDS

3.1.5.7 Port G

Table 3-11 Port G Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
29	PG0	I/O	Z	PU/PD	4 mA	VCC-PG
30	PG1	I/O	Z	PU/PD	4 mA	VCC-PG
31	PG2	I/O	Z	PU/PD	4 mA	VCC-PG
32	PG3	I/O	Z	PU/PD	4 mA	VCC-PG
33	PG4	I/O	Z	PU/PD	4 mA	VCC-PG
34	PG5	I/O	Z	PU/PD	4 mA	VCC-PG
35	PG6	I/O	Z	PU/PD	4 mA	VCC-PG
36	PG7	I/O	Z	PU/PD	4 mA	VCC-PG
38	PG8	I/O	Z	PU/PD	4 mA	VCC-PG
39	PG9	I/O	Z	PU/PD	4 mA	VCC-PG
40	PG10	I/O	Z	PU/PD	4 mA	VCC-PG
41	PG11	I/O	Z	PU/PD	4 mA	VCC-PG
42	PG12	I/O	Z	PU/PD	4 mA	VCC-PG
43	PG13	I/O	Z	PU/PD	4 mA	VCC-PG
44	PG14	I/O	Z	PU/PD	4 mA	VCC-PG

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
45	PG15	I/O	Z	PU/PD	4 mA	VCC-PG
37	VCC-PG	P	N/A	N/A	N/A	N/A

3.1.6 USB

Table 3-12 USB Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
112	USB0-DM	A I/O	N/A	N/A	N/A	VCC33-USB-LDOIN-IO
113	USB0-DP	A I/O	N/A	N/A	N/A	VCC33-USB-LDOIN-IO
110	USB1-DM	A I/O	N/A	N/A	N/A	VCC33-USB-LDOIN-IO
111	USB1-DP	A I/O	N/A	N/A	N/A	VCC33-USB-LDOIN-IO

3.1.7 Audio Codec

Table 3-13 Audio Codec Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
96	MICIN1P	AI	N/A	N/A	N/A	AVCC
97	MICIN1N	AI	N/A	N/A	N/A	AVCC
91	FMINL	AI	N/A	N/A	N/A	AVCC
90	FMINR	AI	N/A	N/A	N/A	AVCC
93	LINEINL	AI	N/A	N/A	N/A	AVCC
92	LINEINR	AI	N/A	N/A	N/A	AVCC
89	LINEOUTLN	AO	N/A	N/A	N/A	AVCC
88	LINEOUTLP	AO	N/A	N/A	N/A	AVCC
87	LINEOUTRN	AO	N/A	N/A	N/A	AVCC
86	LINEOUTRP	AO	N/A	N/A	N/A	AVCC
101	HPOUTL	AO	N/A	N/A	N/A	HPVCC
99	HPOUTR	AO	N/A	N/A	N/A	HPVCC
100	HPOUTFB	A I/O	N/A	N/A	N/A	HPVCC
102	HPVCC	P	N/A	N/A	N/A	N/A
94	VRA1	A I/O	N/A	N/A	N/A	AVCC

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
98	AVCC	P	N/A	N/A	N/A	N/A
95	AGND	G	N/A	N/A	N/A	N/A

3.1.8 GPADC

Table 3-14 GPADC Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
85	GPADC0	AI	N/A	N/A	N/A	AVCC
84	GPADC1	AI	N/A	N/A	N/A	AVCC

3.1.9 Power

Table 3-15 Power Pin Characteristics

Pin# ^[1]	Pin Name ^[2]	Type ^[3]	Ball Reset State ^[4]	Pull Up/Down ^[5]	Default Buffer Strength ^[6]	I/O Power Supply ^[7]
28	VDD-SYS0	P	N/A	N/A	N/A	N/A
65	VDD-SYS1	P	N/A	N/A	N/A	N/A
68	VDD-SYS2	P	N/A	N/A	N/A	N/A
79	VDD-SYS3	P	N/A	N/A	N/A	N/A
121	VDD-SYS4	P	N/A	N/A	N/A	N/A
24	VCC33-USB-LDOIN-IO	P	N/A	N/A	N/A	N/A
19	VCC18-EFUSE	P	N/A	N/A	N/A	N/A

3.2 GPIO Multiplex Function

The following tables provide a description of the GPIO multiplex function.



NOTE

For each GPIO, Function0 is input function; Function1 is output function; Function9 to Function13 are reserved.

3.2.1 Port A

Table 3-16 Port A Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function14
PA0	I/O	TP-X1	PWM-1						PA-EINT0
PA1	I/O	TP-X2	PWM-2						PA-EINT1
PA2	I/O	TP-Y1	PWM-3						PA-EINT2
PA3	I/O	TP-Y2	PWM-4						PA-EINT3

3.2.2 Port B

Table 3-17 Port B Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function14
PB1	I/O	LCD0-D0	I2S2-DOUT3		I2S2-DIN3			IR-RX	PB-EINT1
PB2	I/O	LCD0-D1	I2S2-DOUT2	TWI0-SDA	I2S2-DIN2		UART4-TX		PB-EINT2
PB3	I/O	LCD0-D8	I2S2-DOUT1	TWI0-SCK	I2S2-DIN0		UART4-RX		PB-EINT3
PB4	I/O	LCD0-D9	I2S2-DOUT0	TWI1-SCK	I2S2-DIN1		UART5-TX		PB-EINT4
PB5	I/O	LCD0-D16	I2S2-BCLK	TWI1-SDA	PWM-0		UART5-RX		PB-EINT5
PB6	I/O	LCD0-D17	I2S2-LRCK	TWI3-SCK	PWM-1		UART3-TX		PB-EINT6
PB7	I/O	PWM-4	I2S2-MCLK	TWI3-SDA	IR-RX		UART3-RX		PB-EINT7

3.2.3 Port C

Table 3-18 Port C Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function14
PC0	I/O	SPI0-CLK	SDC2-CLK						PC-EINT0
PC1	I/O	SPI0-CS0	SDC2-CMD						PC-EINT1
PC2	I/O	SPI0-MISO	SDC2-D2	BOOT-SEL0					PC-EINT2
PC3	I/O	SPI0-MOSI	SDC2-D0	BOOT-SEL1					PC-EINT3
PC4	I/O	SPI0-WP	SDC2-D1	UART3-TX	TWI3-SCK				PC-EINT4
PC5	I/O	SPI0-HOLD	SDC2-D3	UART3-RX	TWI3-SDA	TCON-FSYNC			PC-EINT5
PC6	I/O	UART2-TX	SDC2-RST	TWI2-SCK	LEDC				PC-EINT6
PC7	I/O	UART2-RX	PWM-4	TWI2-SDA					PC-EINT7
PC8	I/O		PWM-5	TWI2-SCK		UART0-TX	UART1-TX		PC-EINT8
PC9	I/O		PWM-6	TWI2-SDA		UART0-RX	UART1-RX		PC-EINT9

3.2.4 Port D

Table 3-19 Port D Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function14
PD0	I/O	LCD0-D2	LVDS0-D0P	DSI-D0P	TWI0-SCK				PD-EINT0
PD1	I/O	LCD0-D3	LVDS0-D0N	DSI-D0N	UART2-TX				PD-EINT1
PD2	I/O	LCD0-D4	LVDS0-D1P	DSI-D1P	UART2-RX				PD-EINT2
PD3	I/O	LCD0-D5	LVDS0-D1N	DSI-D1N	UART2-RTS				PD-EINT3

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function14
PD4	I/O	LCD0-D6	LVDS0-D2P	DSI-CKP	UART2-CTS				PD-EINT4
PD5	I/O	LCD0-D7	LVDS0-D2N	DSI-CKN	UART5-TX				PD-EINT5
PD6	I/O	LCD0-D10	LVDS0-CKP	DSI-D2P	UART5-RX				PD-EINT6
PD7	I/O	LCD0-D11	LVDS0-CKN	DSI-D2N	UART4-TX				PD-EINT7
PD8	I/O	LCD0-D12	LVDS0-D3P	DSI-D3P	UART4-RX				PD-EINT8
PD9	I/O	LCD0-D13	LVDS0-D3N	DSI-D3N	PWM-6				PD-EINT9
PD10	I/O	LCD0-D14	LVDS1-D0P	SPI1-CS0/DBI-CSX	UART3-TX				PD-EINT10
PD11	I/O	LCD0-D15	LVDS1-D0N	SPI1-CLK/DBI-SCLK	UART3-RX				PD-EINT11
PD12	I/O	LCD0-D18	LVDS1-D1P	SPI1-MOSI/DBI-SDO	TWI0-SDA				PD-EINT12
PD13	I/O	LCD0-D19	LVDS1-D1N	SPI1-MISO/DBI-SDI/DBI-TE/DBI-DCX	UART3-RTS				PD-EINT13
PD14	I/O	LCD0-D20	LVDS1-D2P	SPI1-HOLD/DBI-DCX/DBI-WRX	UART3-CTS				PD-EINT14
PD15	I/O	LCD0-D21	LVDS1-D2N	SPI1-WP/DBI-TE	IR-RX				PD-EINT15
PD16	I/O	LCD0-D22	LVDS1-CKP	DMIC-DATA3	PWM-0				PD-EINT16
PD17	I/O	LCD0-D23	LVDS1-CKN	DMIC-DATA2	PWM-1				PD-EINT17
PD18	I/O	LCD0-CLK	LVDS1-D3P	DMIC-DATA1	PWM-2				PD-EINT18
PD19	I/O	LCD0-DE	LVDS1-D3N	DMIC-DATA0	PWM-3				PD-EINT19
PD20	I/O	LCD0-HSYNC	TWI2-SCK	DMIC-CLK	PWM-4				PD-EINT20
PD21	I/O	LCD0-VSYNC	TWI2-SDA	UART1-TX	PWM-5				PD-EINT21
PD22	I/O	OWA-OUT	IR-RX	UART1-RX	PWM-7				PD-EINT22

3.2.5 Port E

Table 3-20 Port E Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function14
PE0	I/O	NCSI0-HSYNC	LCD0-D0	TWI1-SCK	UART2-TX	UART4-TX		RGMII0-EPHY-25/50M	PE-EINT0
PE1	I/O	NCSI0-VSYNC	LCD0-D1	TWI1-SDA	UART2-RX	UART4-RX		RGMII0-CLKIN/RMII0-RXER	PE-EINT1
PE2	I/O	NCSI0-MCLK	LCD0-D8	TWI0-SCK	CLK-FANOUT0	UART0-TX		RGMII0-RXCTL/RMII0-CRS-DV	PE-EINT2
PE3	I/O	NCSI0-PCLK	LCD0-D9	TWI0-SDA	CLK-FANOUT1	UART0-RX		RGMII0-MDIO	PE-EINT3
PE4	I/O	NCSI0-D1	LCD0-D16	TWI2-SCK	CLK-FANOUT2		RJTAG-MS	RGMII0-MDC	PE-EINT4
PE5	I/O	NCSI0-D7	LCD0-D17	TWI2-SDA	LEDC		RJTAG-DI	RGMII0-TXCTL/RMII0-TXEN	PE-EINT5
PE6	I/O	NCSI0-D6	UART2-TX	TWI3-SCK	OWA-IN		RJTAG-DO	RGMII0-TXD1/RMII0-TXD1	PE-EINT6

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function14
PE7	I/O	NCSI0-D0	UART2-RX	TWI3-SDA	OWA-OUT		RJTAG-CK	RGMII0-TXD0/RMII0-TXD0	PE-EINT7
PE8	I/O	NCSI0-D5	UART2-RTS	PWM-2	UART3-TX			RGMII0-TXCK/RMII0-TXCK	PE-EINT8
PE9	I/O	PWM-5	UART2-CTS	PWM-3	UART3-RX		TWI2-SDA	RGMII0-RXD1/RMII0-RXD1	PE-EINT9
PE10	I/O	NCSI0-D4	UART4-TX	PWM-4	IR-RX			RGMII0-RXD0/RMII0-RXD0	PE-EINT10
PE11	I/O	NCSI0-FIELD	UART4-RX	IR-TX	I2S0-MCLK	DMIC-CLK	TWI2-SCK	RGMII0-RXCK/RMII0-NULL	PE-EINT11
PE12	I/O	NCSI0-D3	UART5-TX	PWM-7	I2S0-BCLK	DMIC-DATA0	LCD0-HSYNC	RGMII0-RXD2/RMII0-NULL	PE-EINT12
PE13	I/O	NCSI0-D2	UART5-RX	PWM-6	I2S0-LRCK	DMIC-DATA1	LCD0-VSYNC	RGMII0-RXD3/RMII0-NULL	PE-EINT13
PE14	I/O	TWI1-SCK	UART1-TX	I2S0-DOUT1	I2S0-DIN0	DMIC-DATA2	UART2-TX	RGMII0-TXD2/RMII0-NULL	PE-EINT14
PE15	I/O	TWI1-SDA	UART1-RX	I2S0-DOUT0	I2S0-DIN1	DMIC-DATA3	UART2-RX	RGMII0-TXD3/RMII0-NULL	PE-EINT15
PE16	I/O			I2S0-DOUT2	I2S0-DIN2				PE-EINT16

3.2.6 Port F

Table 3-21 Port F Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function14
PF0	I/O	SDC0-D1	DJTAG-MS	RJTAG-MS	I2S2-DOUT1	I2S2-DIN0	LCD0-D0		PF-EINT0
PF1	I/O	SDC0-D0	DJTAG-DI	RJTAG-DI	I2S2-DOUT0	I2S2-DIN1	LCD0-D1		PF-EINT1
PF2	I/O	SDC0-CLK	UART0-TX	TWI0-SCK	LEDC	OWA-IN	LCD0-D8		PF-EINT2
PF3	I/O	SDC0-CMD	DJTAG-DO	RJTAG-DO	I2S2-BCLK	TWI1-SCK	LCD0-D9		PF-EINT3
PF4	I/O	SDC0-D3	UART0-RX	TWI0-SDA	PWM-6	IR-TX	LCD0-D16		PF-EINT4
PF5	I/O	SDC0-D2	DJTAG-CK	RJTAG-CK	I2S2-LRCK	TWI1-SDA	LCD0-D17		PF-EINT5
PF6	I/O		OWA-OUT	IR-RX	I2S2-MCLK	PWM-5			PF-EINT6

3.2.7 Port G

Table 3-22 Port G Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function14
PG0	I/O	SDC1-D1	UART3-TX	RGMII1-EPHY-25/50M	PWM-7	SDC1-D3	SDC1-D1		PG-EINT0
PG1	I/O	SDC1-D0	UART3-RX	RGMII1-CLKIN/RMII1-RXER	PWM-6	SDC1-D1	SDC1-D3		PG-EINT1
PG2	I/O	SDC1-CLK	UART3-RTS	RGMII1-RXCTL/RMII1-CRS-DV	UART4-TX	SDC1-CLK	SDC1-CLK		PG-EINT2
PG3	I/O	SDC1-CMD	UART3-CTS	RGMII1-MDIO	UART4-RX	SDC1-D0	SDC1-D2		PG-EINT3
PG4	I/O	SDC1-D3	UART5-TX	RGMII1-MDC	PWM-5	SDC1-CMD	SDC1-CMD		PG-EINT4



Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function14
PG5	I/O	SDC1-D2	UART5-RX	RGMII1-TXCTL/RMII1-TXEN	PWM-4	SDC1-D2	SDC1-D0		PG-EINT5
PG6	I/O	UART1-TX	TWI2-SCK	RGMII1-TXD1/RMII1-TXD1	PWM-1				PG-EINT6
PG7	I/O	UART1-RX	TWI2-SDA	RGMII1-TXD0/RMII1-TXD0	OWA-IN				PG-EINT7
PG8	I/O	UART1-RTS	TWI1-SCK	RGMII1-TXCK/RMII1-TXCK	UART3-TX				PG-EINT8
PG9	I/O	UART1-CTS	TWI1-SDA	RGMII1-RXD1/RMII1-RXD1	UART3-RX				PG-EINT9
PG10	I/O	PWM-3	TWI3-SCK	RGMII1-RXD0/RMII1-RXD0	CLK-FANOUT0	IR-RX			PG-EINT10
PG11	I/O	I2S1-MCLK	TWI3-SDA	RGMII1-RXCK/RMII1-NULL	CLK-FANOUT1	TCON-FSYNC			PG-EINT11
PG12	I/O	I2S1-LRCK	TWI0-SCK	RGMII1-RXD2/RMII1-NULL	CLK-FANOUT2	PWM-0	UART1-TX		PG-EINT12
PG13	I/O	I2S1-BCLK	TWI0-SDA	RGMII1-RXD3/RMII1-NULL	PWM-2	LEDC	UART1-RX		PG-EINT13
PG14	I/O	I2S1-DIN0	TWI2-SCK	RGMII1-TXD2/RMII1-NULL	I2S1-DOUT1		UART1-RTS		PG-EINT14
PG15	I/O	I2S1-DOUT0	TWI2-SDA	RGMII1-TXD3/RMII1-NULL	I2S1-DIN1		UART1-CTS		PG-EINT15

3.3 Detailed Signal Description

The following tables show the detailed function description of every signal based on the different interfaces.

[1] **Signal Name:** The name of every signal.

[2] **Description:** The detailed function description of every signal.

[3] **Type:** Denotes the signal direction:

I (Input),

O (Output),

I/O (Input/Output),

OD (Open-Drain),

A (Analog),

AI (Analog Input),

AO (Analog Output),

A I/O (Analog Input/Output),

P (Power),

G (Ground)

3.3.1 SDRAM

Table 3-23 SDRAM Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
VCC-DRAM[1:0]	DRAM IO Power Supply	P
VDD18-DRAM	SDRAM Controller Power Supply	P

3.3.2 System Control

Table 3-24 System Control Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
RESET	SoC Reset Signal (Active Low)	I/O, OD
TEST	Test Signal	I
BOOT-SEL[1:0]	Boot Media Select	I

3.3.3 RTC&PLL

Table 3-25 RTC&PLL Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
NMI	Non-Maskable Interrupt	I/O, OD
VCC-RTC-PLL	PLL/RTC/DCXO/LDO-OUT Power Supply	P

3.3.4 DCXO

Table 3-26 DCXO Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
DXIN	Digital Compensated Crystal Oscillator Input	AI
DXOUT	Digital Compensated Crystal Oscillator Output	AO
REFCLK	Digital Compensated Crystal Oscillator Clock Fanout	AO

3.3.5 Clock Fanout

Table 3-27 Clock Fanout Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
CLK-FANOUT0	Internal Clock Fanout Optional Frequency: 32 kHz, 12 MHz, 16 MHz, 24 MHz, 25 MHz, 27 MHz, and so on	O
CLK-FANOUT1	Internal Clock Fanout Optional Frequency: 32 kHz, 12 MHz, 16 MHz, 24 MHz, 25 MHz, 27 MHz, and so on	O
CLK-FANOUT2	Internal Clock Fanout Optional Frequency: 32 kHz, 12 MHz, 16 MHz, 24 MHz, 25 MHz, 27 MHz, and so on	O

3.3.6 SD Card/SDIO/eMMC

Table 3-28 SD Card/SDIO/eMMC Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
SDC0-CMD	SD Card Command Output/Response Input	I/O, OD
SDC0-CLK	SD Card Clock Output	O
SDC0-D0	SD Card Data Input/ Output 0	I/O
SDC0-D1	SD Card Data Input/ Output 1	I/O

Signal Name ^[1]	Description ^[2]	Type ^[3]
SDC0-D2	SD Card Data Input/ Output 2	I/O
SDC0-D3	SD Card Data Input/ Output 3	I/O
SDC1-CMD	SDIO WIFI Command Output/Response Input	I/O, OD
SDC1-CLK	SDIO WIFI Clock Output	O
SDC1-D0	SDIO WIFI Data Input/ Output 0	I/O
SDC1-D1	SDIO WIFI Data Input/ Output 1	I/O
SDC1-D2	SDIO WIFI Data Input/ Output 2	I/O
SDC1-D3	SDIO WIFI Data Input/ Output 3	I/O
SDC2-CMD	eMMC Command Output/Response Input	I/O, OD
SDC2-CLK	eMMC Clock Output	O
SDC2-D0	eMMC Data Input/ Output 0	I/O
SDC2-D1	eMMC Data Input/ Output 1	I/O
SDC2-D2	eMMC Data Input/ Output 2	I/O
SDC2-D3	eMMC Data Input/ Output 3	I/O
SDC2-RST	eMMC Reset	O

3.3.7 USB

Table 3-29 USB Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
USB0-DM	USB0 Differential Data (Negative)	A I/O
USB0-DP	USB0 Differential Data (Positive)	A I/O
USB1-DM	USB1 Differential Data (Negative)	A I/O
USB1-DP	USB1 Differential Data (Positive)	A I/O

3.3.8 I2S/PCM

Table 3-30 I2S/PCM Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
I2S0-MCLK	I2S0 Master Clock	O
I2S0-LRCK	I2S0/PCM0 Sample Rate Clock/Sync	I/O
I2S0-BCLK	I2S0/PCM0 Bit Rate Clock	I/O
I2S0-DIN0	I2S0/PCM0 Serial Data Input 0	I

Signal Name ^[1]	Description ^[2]	Type ^[3]
I2S0-DIN1	I2S0/PCM0 Serial Data Input 1	I
I2S0-DIN2	I2S0/PCM0 Serial Data Input 2	I
I2S0-DOUT0	I2S0/PCM0 Serial Data Output 0	O
I2S0-DOUT1	I2S0/PCM0 Serial Data Output 1	O
I2S0-DOUT2	I2S0/PCM0 Serial Data Output 2	O
I2S1-MCLK	I2S1 Master Clock	O
I2S1-LRCK	I2S1/PCM1 Sample Rate Clock/Sync	I/O
I2S1-BCLK	I2S1/PCM1 Bit Rate Clock	I/O
I2S1-DIN0	I2S1/PCM1 Serial Data Input 0	I
I2S1-DIN1	I2S1/PCM1 Serial Data Input 1	I
I2S1-DOUT0	I2S1/PCM1 Serial Data Output 0	O
I2S1-DOUT1	I2S1/PCM1 Serial Data Output 1	O
I2S2-MCLK	I2S2 Master Clock	O
I2S2-LRCK	I2S2/PCM2 Sample Rate Clock/Sync	I/O
I2S2-BCLK	I2S2/PCM2 Bit Rate Clock	I/O
I2S2-DIN0	I2S2/PCM2 Serial Data Input 0	I
I2S2-DIN1	I2S2/PCM2 Serial Data Input 1	I
I2S2-DIN2	I2S2/PCM2 Serial Data Input 2	I
I2S2-DIN3	I2S2/PCM2 Serial Data Input 3	I
I2S2-DOUT0	I2S2/PCM2 Serial Data Output 0	O
I2S2-DOUT1	I2S2/PCM2 Serial Data Output 1	O
I2S2-DOUT2	I2S2/PCM2 Serial Data Output 2	O
I2S2-DOUT3	I2S2/PCM2 Serial Data Output 3	O

3.3.9 Audio Codec

Table 3-31 Audio Codec Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
MICIN1P	Microphone Differential Positive Input 1	AI
MICIN1N	Microphone Differential Negative Input 1	AI
FMINL	FMIN Left Input	AI
FMINR	FMIN Right Input	AI

Signal Name ^[1]	Description ^[2]	Type ^[3]
LINEINL	LINEIN Left Single-End Input	AI
LINEINR	LINEIN Right Single-End Input	AI
LINEOUTLN	Lineout Left Channel Negative Differential Output	AO
LINEOUTLP	Lineout Left Channel Positive Differential Output	AO
LINEOUTRN	Lineout Right Channel Negative Differential Output	AO
LINEOUTRP	Lineout Right Channel Positive Differential Output	AO
HPOUTFB	Pseudo Differential Headphone Ground Reference	A I/O
HPOUTL	Headphone Left Output	AO
HPOUTR	Headphone Right Output	AO
HPVCC	Headphone Power	P
AVCC	Power Supply for Analog Part	P
VRA1	Internal Reference Voltage	A I/O
AGND	Analog Ground	G

3.3.10 DMIC

Table 3-32 DMIC Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
DMIC-CLK	Digital Microphone Clock Output	O
DMIC-DATA0	Digital Microphone Data Input 0	I
DMIC-DATA1	Digital Microphone Data Input 1	I
DMIC-DATA2	Digital Microphone Data Input 2	I
DMIC-DATA3	Digital Microphone Data Input 3	I

3.3.11 OWA

Table 3-33 OWA Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
OWA-IN	One Wire Audio Input	I
OWA-OUT	One Wire Audio Output	O

3.3.12 GPADC

Table 3-34 GPADC Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
GPADC0	General Purpose ADC Input 0	AI
GPADC1	General Purpose ADC Input 1	AI

3.3.13 TPADC

Table 3-35 TPADC Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
TP-X1	Touch Panel X1 Input	AI
TP-X2	Touch Panel X2 Input	AI
TP-Y1	Touch Panel Y1 Input	AI
TP-Y2	Touch Panel Y2 Input	AI

3.3.14 Parallel CSI

Table 3-36 Parallel CSI Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
NCSI0-PCLK	Parallel CSI Pixel Clock Input	I
NCSI0-MCLK	Parallel CSI Master Clock Output	O
NCSI0-HSYNC	Parallel CSI Horizontal Synchronous Input	I
NCSI0-VSYNC	Parallel CSI Vertical Synchronous Input	I
NCSI0-FIELD	Parallel CSI Field Index	I
NCSI0-D0	Parallel CSI Pixel Data 0	I
NCSI0-D1	Parallel CSI Pixel Data 1	I
NCSI0-D2	Parallel CSI Pixel Data 2	I
NCSI0-D3	Parallel CSI Pixel Data 3	I
NCSI0-D4	Parallel CSI Pixel Data 4	I
NCSI0-D5	Parallel CSI Pixel Data 5	I
NCSI0-D6	Parallel CSI Pixel Data 6	I
NCSI0-D7	Parallel CSI Pixel Data 7	I

3.3.15 LCD

Table 3-37 LCD Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
LCD0-CLK	LCD Clock	O
LCD0-VSYNC	LCD Vertical Synchronization	O
LCD0-HSYNC	LCD Horizontal Synchronization	O
LCD0-DE	LCD Data Enable	O
TCON-FSYNC	Frame Synchronization Signal for TCON and Sensor	O
LCD0-D0	LCD Data Input/Output 0	I/O
LCD0-D1	LCD Data Input/Output 1	I/O
LCD0-D2	LCD Data Input/Output 2	I/O
LCD0-D3	LCD Data Input/Output 3	I/O
LCD0-D4	LCD Data Input/Output 4	I/O
LCD0-D5	LCD Data Input/Output 5	I/O
LCD0-D6	LCD Data Input/Output 6	I/O
LCD0-D7	LCD Data Input/Output 7	I/O
LCD0-D8	LCD Data Input/Output 8	I/O
LCD0-D9	LCD Data Input/Output 9	I/O
LCD0-D10	LCD Data Input/Output 10	I/O
LCD0-D11	LCD Data Input/Output 11	I/O
LCD0-D12	LCD Data Input/Output 12	I/O
LCD0-D13	LCD Data Input/Output 13	I/O
LCD0-D14	LCD Data Input/Output 14	I/O
LCD0-D15	LCD Data Input/Output 15	I/O
LCD0-D16	LCD Data Input/Output 16	I/O
LCD0-D17	LCD Data Input/Output 17	I/O
LCD0-D18	LCD Data Input/Output 18	I/O
LCD0-D19	LCD Data Input/Output 19	I/O
LCD0-D20	LCD Data Input/Output 20	I/O
LCD0-D21	LCD Data Input/Output 21	I/O
LCD0-D22	LCD Data Input/Output 22	I/O

Signal Name ^[1]	Description ^[2]	Type ^[3]
LCD0-D23	LCD0 Data Input/Output 23	I/O

3.3.16 LVDS

Table 3-38 LVDS Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
LVDS0-CKP	LVDS0 Differential Clock (Positive)	AO
LVDS0-CKN	LVDS0 Differential Clock (Negative)	AO
LVDS0-D0P	LVDS0 Differential Data 0 (Positive)	AO
LVDS0-D0N	LVDS0 Differential Data 0 (Negative)	AO
LVDS0-D1P	LVDS0 Differential Data 1 (Positive)	AO
LVDS0-D1N	LVDS0 Differential Data 1 (Negative)	AO
LVDS0-D2P	LVDS0 Differential Data 2 (Positive)	AO
LVDS0-D2N	LVDS0 Differential Data 2 (Negative)	AO
LVDS0-D3P	LVDS0 Differential Data 3 (Positive)	AO
LVDS0-D3N	LVDS0 Differential Data 3 (Negative)	AO
LVDS1-CKP	LVDS1 Differential Clock (Positive)	AO
LVDS1-CKN	LVDS1 Differential Clock (Negative)	AO
LVDS1-D0P	LVDS1 Differential Data 0 (Positive)	AO
LVDS1-D0N	LVDS1 Differential Data 0 (Negative)	AO
LVDS1-D1P	LVDS1 Differential Data 1 (Positive)	AO
LVDS1-D1N	LVDS1 Differential Data 1 (Negative)	AO
LVDS1-D2P	LVDS1 Differential Data 2 (Positive)	AO
LVDS1-D2N	LVDS1 Differential Data 2 (Negative)	AO
LVDS1-D3P	LVDS1 Differential Data 3 (Positive)	AO
LVDS1-D3N	LVDS1 Differential Data 3 (Negative)	AO
VCC18-LVDS	Power Supply for LVDS0/1	P

3.3.17 MIPI DSI

Table 3-39 MIPI DSI Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
DSI-CKP	DSI Differential Clock (Positive)	AO

Signal Name ^[1]	Description ^[2]	Type ^[3]
DSI-CKN	DSI Differential Clock (Negative)	AO
DSI-D0P	DSI Differential Data Line 0 (Positive)	A I/O
DSI-D0N	DSI Differential Data Line 0 (Negative)	A I/O
DSI-D1P	DSI Differential Data Line 1 (Positive)	AO
DSI-D1N	DSI Differential Data Line 1 (Negative)	AO
DSI-D2P	DSI Differential Data Line 2 (Positive)	AO
DSI-D2N	DSI Differential Data Line 2 (Negative)	AO
DSI-D3P	DSI Differential Data Line 3 (Positive)	AO
DSI-D3N	DSI Differential Data Line 3 (Negative)	AO

3.3.18 GMAC

Table 3-40 GMAC Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
RGMIIO-RXCK/ RMII0-NUL	RGMIIO Receive Clock	I
RGMIIO-TXCK/RMII0-TXCK	RGMIIO/RMII0 Transmit Clock For RGMII, IO type is output; For RMII, RMII 50 MHz Reference Clock input, IO type is input.	I/O
RGMIIO-CLKIN/RMII0-RXER	RGMIIO Transmit Clock from External/RMII0 Receive Error	I
RGMIIO-RXCTL/RMII0-CRS-DV	RGMIIO Receive Control/RMII0 Carrier Sense Receive Data Valid	I
RGMIIO-TXCTL/RMII0-TXEN	RGMIIO Transmit Control/RMII0 Transmit Enable	O
RGMIIO-MDC	RGMIIO Management Data Clock	O
RGMIIO-MDIO	RGMIIO Management Data Input/ Output	I/O
RGMIIO-EPHY-25/50M	GMAC PHY 25 MHz or 50 MHz Clock Output	O
RGMIIO-RXD0/RMII0-RXD0	RGMIIO/RMII0 Receive Data 0	I
RGMIIO-RXD1/RMII0-RXD1	RGMIIO/RMII0 Receive Data 1	I
RGMIIO-RXD2/RMII0-NUL	RGMIIO Receive Data 2	I
RGMIIO-RXD3/RMII0-NUL	RGMIIO Receive Data 3	I
RGMIIO-TXD0/RMII0-TXD0	RGMIIO/RMII0 Transmit Data 0	O
RGMIIO-TXD1/RMII0-TXD1	RGMIIO/RMII0 Transmit Data 1	O

Signal Name ^[1]	Description ^[2]	Type ^[3]
RGMII0-TXD2/RMII0-NULL	RGMII0 Transmit Data 2	O
RGMII0-TXD3/RMII0-NULL	RGMII0 Transmit Data 3	O
RGMII1-RXCK/RMII1-NULL	RGMII1 Receive Clock	I
RGMII1-TXCK/RMII1-TXCK	RGMII1/RMII1 Transmit Clock For RGMII, IO type is output; For RMII, RMII 50 MHz Reference Clock input, IO type is input.	I/O
RGMII1-CLKIN/RMII1-RXER	RGMII1 Transmit Clock from External/RMII1 Receive Error	I
RGMII1-RXCTL/RMII1-CRS-DV	RGMII1 Receive Control/RMII1 Carrier Sense Receive Data Valid	I
RGMII1-TXCTL/RMII1-TXEN	RGMII1 Transmit Control/RMII1 Transmit Enable	O
RGMII1-MDC	RGMII1 Management Data Clock	O
RGMII1-MDIO	RGMII1 Management Data Input/ Output	I/O
RGMII1-EPHY-25/50M	GMAC PHY 25 MHz or 50 MHz Clock Output	O
RGMII1-RXD0/RMII1-RXD0	RGMII1/RMII1 Receive Data 0	I
RGMII1-RXD1/RMII1-RXD1	RGMII1/RMII1 Receive Data 1	I
RGMII1-RXD2/RMII1-NULL	RGMII1 Receive Data 2	I
RGMII1-RXD3/RMII1-NULL	RGMII1 Receive Data 3	I
RGMII1-TXD0/RMII1-TXD0	RGMII1/RMII1 Transmit Data 0	O
RGMII1-TXD1/RMII1-TXD1	RGMII1/RMII1 Transmit Data 1	O
RGMII1-TXD2/RMII1-NULL	RGMII1 Transmit Data 2	O
RGMII1-TXD3/RMII1-NULL	RGMII1 Transmit Data 3	O

3.3.19 SPI&SPI DBI

Table 3-41 SPI&SPI DBI Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
SPI0-CS0	SPI0 Chip Select 0 (Active Low)	I/O
SPI0-CLK	SPI0 Clock	I/O
SPI0-MOSI	SPI0 Master Data Out, Slave Data In	I/O
SPI0-MISO	SPI0 Master Data In, Slave Data Out	I/O

Signal Name ^[1]	Description ^[2]	Type ^[3]
SPI0-WP	SPI0 Write Protection (Active Low)/ Serial Data Input and Output for Quad Input or Quad Output	I/O
SPI0-HOLD	SPI0 Hold Signal/ Serial Data Input and Output for Quad Input or Quad Output	I/O
SPI1-CS0	SPI1 Chip Select 0 (Active Low)	I/O
SPI1-CLK	SPI1 Clock	I/O
SPI1-MOSI	SPI1 Master Data Out, Slave Data In	I/O
SPI1-MISO	SPI1 Master Data In, Slave Data Out	I/O
SPI1-WP	SPI1 Write Protection (Active Low)/ Serial Data Input and Output for Quad Input or Quad Output	I/O
SPI1-HOLD	SPI1 Hold Signal/ Serial Data Input and Output for Quad Input or Quad Output	I/O
DBI-CSX	Chip Select Signal (Active Low)	I/O
DBI-SCLK	Serial Clock Signal	I/O
DBI-SDO	Data Output Signal	I/O
DBI-SDI	Data Input Signal The data is sampled on the rising edge and the falling edge.	I/O
DBI-TE	Tearing Effect Input It is used to capture the external TE signal edge. The rising and falling edge is configurable.	I/O
DBI-DCX	DCX pin is the select output signal of data and command. DCX = 0: register command; DCX = 1: data or parameter.	I/O
DBI-WRX	When DBI operates in dual data lane format, the RGB666 format 2 can use WRX to transfer data	I/O

3.3.20 UART

Table 3-42 UART Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
UART0-RX	UART0 Data Receiver	I
UART0-TX	UART0 Data Transmitter	O
UART1-CTS	UART1 Clear to Send	I
UART1-RTS	UART1 Request to Send	O

Signal Name ^[1]	Description ^[2]	Type ^[3]
UART1-RX	UART1 Data Receiver	I
UART1-TX	UART1 Data Transmitter	O
UART2-CTS	UART2 Clear to Send	I
UART2-RTS	UART2 Request to Send	O
UART2-RX	UART2 Data Receiver	I
UART2-TX	UART2 Data Transmitter	O
UART3-CTS	UART3 Clear to Send	I
UART3-RTS	UART3 Request to Send	O
UART3-RX	UART3 Data Receiver	I
UART3-TX	UART3 Data Transmitter	O
UART4-RX	UART4 Data Receiver	I
UART4-TX	UART4 Data Transmitter	O
UART5-RX	UART5 Data Receiver	I
UART5-TX	UART5 Data Transmitter	O

3.3.21 PWM

Table 3-43 PWM Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
PWM-0	PWM Wave Output /Capture Wave Input 0	I/O
PWM-1	PWM Wave Output /Capture Wave Input 1	I/O
PWM-2	PWM Wave Output /Capture Wave Input 2	I/O
PWM-3	PWM Wave Output /Capture Wave Input 3	I/O
PWM-4	PWM Wave Output /Capture Wave Input 4	I/O
PWM-5	PWM Wave Output /Capture Wave Input 5	I/O
PWM-6	PWM Wave Output /Capture Wave Input 6	I/O
PWM-7	PWM Wave Output /Capture Wave Input 7	I/O

3.3.22 TWI

Table 3-44 TWI Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
TWI0-SCK	TWI0 Serial Clock Signal	I/O

Signal Name ^[1]	Description ^[2]	Type ^[3]
TWI0-SDA	TWI0 Serial Data Signal	I/O
TWI1-SCK	TWI1 Serial Clock Signal	I/O
TWI1-SDA	TWI1 Serial Data Signal	I/O
TWI2-SCK	TWI2 Serial Clock Signal	I/O
TWI2-SDA	TWI2 Serial Data Signal	I/O
TWI3-SCK	TWI3 Serial Clock Signal	I/O
TWI3-SDA	TWI3 Serial Data Signal	I/O

3.3.23 IR RX

Table 3-45 IR RX Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
IR-RX	Consumer Infrared Receiver	I

3.3.24 IR TX

Table 3-46 IR TX Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
IR-TX	Consumer Infrared Transmitter	O

3.3.25 LEDC

Table 3-47 LEDC Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
LEDC	Intelligent Control LED Signal Output	O

3.3.26 JTAG

Table 3-48 JTAG Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
RJTAG-MS	RISC-V JTAG Mode Selection	I
RJTAG-CK	RISC-V JTAG Clock Signal	I
RJTAG-DI	RISC-V JTAG Data Input	I
RJTAG-DO	RISC-V JTAG Data Output	O
DJTAG-MS	DAP JTAG Mode Selection	I

Signal Name ^[1]	Description ^[2]	Type ^[3]
DJTAG-CK	DAP JTAG Clock Signal	I
DJTAG-DI	DAP JTAG Data Input	I
DJTAG-DO	DAP JTAG Data Output	O

3.3.27 Interrupt

Table 3-49 Interrupt Signal Description

Signal Name ^[1]	Description ^[2]	Type ^[3]
PA-EINT[3:0]	Port A Interrupt	I
PC-EINT[12:0]	Port C Interrupt	I
PD-EINT[19:0]	Port D Interrupt	I
PE-EINT[15:0]	Port E Interrupt	I
PF-EINT[5:0]	Port F Interrupt	I
PG-EINT[12:0]	Port G Interrupt	I

4 Electrical characteristics

4.1 Parameter Conditions

4.1.1 Minimum and Maximum Values

Unless otherwise specified the minimum and maximum values are guaranteed in the worst conditions of ambient temperature, supply voltage, and frequencies by tests in production on 100% of the devices with ambient temperature at $T_a = 25^\circ\text{C}$ and $T_a = T_a \text{ max}$.

Data based on characterization results, design simulation, and/or technology characteristics are indicated in the table footnotes and are not tested in production.

4.1.2 Typical Values

Unless otherwise specified, the typical data are based on $T_a = 25^\circ\text{C}$. They are given only as design guidelines.

4.1.3 Temperature Definitions

- Ambient Temperature— the temperature of the surrounding environment.
- Junction Temperature— the hottest temperature of the silicon chip inside the package.
- Absolute Maximum Junction Temperature— the temperature beyond which damage occurs to the device. The device may not function or meet expected performance at this temperature.
- Recommended Operating Temperature— the junction temperature at which the device operates continuously at the designated performance over the designed lifetime. The reliability of the device may be degraded if the device operates above this temperature. Some devices will not function electrically above this temperature.

4.2 Absolute Maximum Ratings

Absolute Maximum Ratings are those values beyond which damage to the device may occur. The following table specifies the absolute maximum ratings.



CAUTION

Stresses beyond those listed under Table 4-1 may affect reliability or cause permanent damage to the device. These are stress ratings only. Functional operation of the device at these or any other conditions beyond those indicated under section 4.3 Recommended Operating Conditions is not implied. Exposure to absolute maximum rated conditions for extended periods may affect device reliability.

Table 4-1 Absolute Maximum Ratings

Supply Name	Description	Min ⁽¹⁾	Max ⁽¹⁾	Unit
AVCC	Audio Codec/PA/GPADC Power Supply	-0.3	2.16	V
VCC-RTC-PLL	PLL/RTC/DCXO/LDO-OUT Power Supply	-0.3	2.16	V
HPVCC	Headphone Power Supply	-0.3	2.16	V
VCC18-LVDS	LVDS/PF Power Supply	-0.3	2.16	V
VCC18-EFUSE	EFUSE Power Supply	-0.3	2.16	V
VCC33-USB-LDOIN-IO	PB/PC/USB/LDO-IN Power Supply	-0.3	3.96	V
VCC33-PD	Digital Port D Power	-0.3	3.96	V
VCC-PE	Digital Port F Power	-0.3	3.96	V
VCC-PG	Digital Port G Power	-0.3	3.96	V
VDD-SYS0 VDD-SYS1 VDD-SYS2 VDD-SYS3 VDD-SYS4	CPU/SYS Power Supply	-0.3	1.05	V
VCC-DRAM0 VCC-DRAM1	DRAM Power Supply	-0.3	VDDQ+0.3	V
VDD18-DRAM	DRAM Power Supply	-0.3	2.16	V
V _{ESD} ⁽²⁾	Human Body Model (HBM) ⁽³⁾	-2000	2000	V
	Charged Device Model (CDM) ⁽⁴⁾	-250	250	V
I _{Latch-up}	Latch-up I-test performance current-pulse injection on each IO pin ⁽⁵⁾	Pass		
	Latch-up over-voltage performance voltage injection on each IO pin ⁽⁶⁾	Pass		

- (1) The min/max voltages of power rails are guaranteed by design, not tested in production.
- (2) Electrostatic discharge (ESD) to measure device sensitivity/immunity to damage caused by electrostatic discharges into the devices.
- (3) Level listed above is the passing level per ESDA/JEDEC JS-001-2017.
- (4) Level listed above is the passing level per ESDA/JEDEC JS-002-2018.
- (5) Based on JESD78E; each device is tested with IO pin injection of ± 150 mA at room temperature.

- (6) Based on JESD78E; each device is tested with a stress voltage of $1.5 \times V_{ddmax}$ at room temperature.

4.3 Recommended Operating Conditions

The following table describes operating conditions of the device.



NOTE

Logic functions and parameter values are not assured out of the range specified in the recommended operating conditions.

Table 4-2 Recommended Operating Conditions

Supply Name	Description	Min	Typ	Max	Unit
AVCC	Audio Codec/PA/GPADC Power Supply	1.782	1.8	1.818	V
VCC-RTC-PLL	PLL/RTC/DCXO/LDO-OUT Power Supply	1.764	1.8	1.836	V
HPVCC	Headphone Power Supply	1.764	1.8	1.836	V
VCC18-LVDS	LVDS/PF Power Supply	1.71	1.8	1.89	V
VCC18-EFUSE	EFUSE Power Supply	1.71	1.8	1.89	V
VCC33-USB-LDOIN-IO	PB/PC/USB/LDO-IN Power Supply	2.97	3.3	3.63	V
VCC33-PD	Digital Port D Power 3.3 V Voltage	2.97	3.3	3.63	V
VCC-PE	Digital Port E Power 3.3 V Voltage 1.8 V Voltage	1.62 2.97	1.8 3.3	1.98 3.63	V
VCC-PG	Digital Port G Power 3.3 V Voltage 1.8 V Voltage	1.62 2.97	1.8 3.3	1.98 3.63	V
VDD-SYS0 VDD-SYS1 VDD-SYS2 VDD-SYS3 VDD-SYS4	CPU/SYS Power Supply	0.9	0.98	1.02	V
VCC-DRAM0 VCC-DRAM1	DRAM Power Supply	1.746	1.8	1.854	V

Supply Name	Description	Min	Typ	Max	Unit
VDD18-DRAM	DRAM 1.8V Power Supply	1.7	1.8	1.95	V

4.4 GPIO DC Electrical Characteristics

Table 4-3 summarizes the GPIO DC electrical characteristics of the device.

Table 4-3 GPIO DC Electrical Characteristics⁽¹⁾

(VCC: VCC33-PD/VCC-PE/VCC-PG/AVCC/ VCC33-USB-LDOIN-IO/VCC18-LVDS)

Symbol	Parameter		Min	Typ	Max	Unit
V _{IH}	High-Level Input Voltage		0.7 * VCC	-	1.1 * VCC	V
V _{IL}	Low-Level Input Voltage		-0.3	-	0.25 * VCC	V
R _{PU}	Output Pull-up Resistance	PC1 to PC5, PF3	9	15	21	kΩ
		PG0, PG1, PG3, PG4, PG5	19.8	33	46.2	kΩ
		Other GPIOs	60	100	140	kΩ
R _{PD}	Output Pull-down Resistance	PC1 to PC5, PF3	9	15	21	kΩ
		PG0, PG1, PG3, PG4, PG5	19.8	33	46.2	kΩ
		Other GPIOs	60	100	140	kΩ
I _{IH}	High-Level Input Current		-	-	10	uA
I _{IL}	Low-Level Input Current		-	-	10	uA
V _{OH}	High-Level Output Voltage		VCC – 0.3	-	VCC	V
V _{OL}	Low-Level Output Voltage		0	-	0.2	V
I _{OZ}	Tri-State Output Leakage Current		-10	-	10	uA
C _{IN}	Input Capacitance		-	-	5	pF
C _{OUT}	Output Capacitance		-	-	5	pF

(1) Guaranteed by design.

4.5 SMHC Electrical Characteristics

The SMHC electrical parameters are related to different supply voltage.

Figure 4-1 SMHC Voltage Waveform

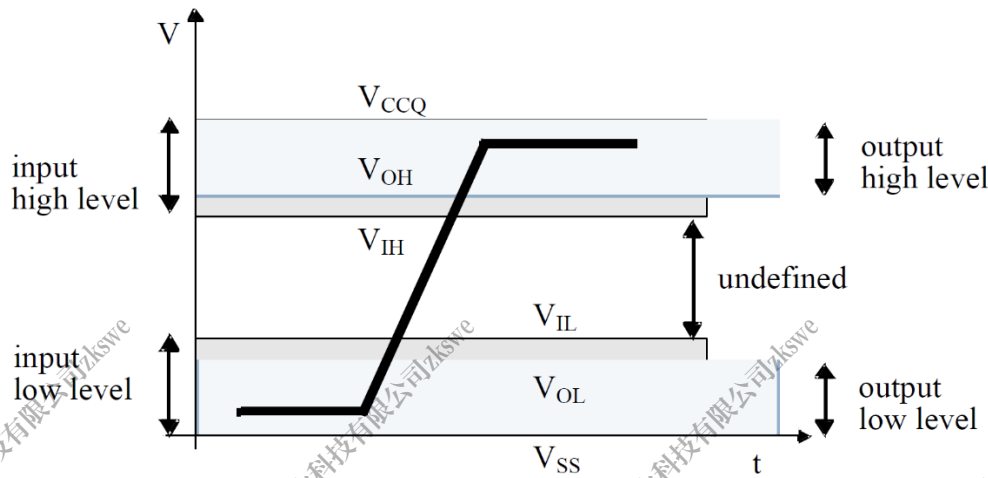


Table 4-4 shows 3.3 V SMHC electrical parameters.

Table 4-4 3.3 V SMHC Electrical Parameters

Symbol	Parameter	Min	Typ	Max	Unit
VDD	Power voltage	2.7	-	3.6	V
V _{CCQ}	I/O voltage	2.7	-	3.6	V
V _{OH}	Output high-level voltage	0.75 * V _{CCQ}	-	-	V
V _{OL}	Output low-level voltage	-	-	0.125 * V _{CCQ}	V
V _{IH}	Input high-level voltage	0.625 * V _{CCQ}	-	V _{CCQ} + 0.3	V
V _{IL}	Input low-level voltage	V _{SS} - 0.3	-	0.25 * V _{CCQ}	V

Table 4-5 shows 1.8 V SMHC electrical parameters.

Table 4-5 1.8 V SMHC Electrical Parameters

Symbol	Parameter	Min	Typ	Max	Unit
VDD	Power voltage	2.7	-	3.6	V
V _{CCQ}	I/O voltage	1.7	-	1.95	V
V _{OH}	Output high-level voltage	V _{CCQ} - 0.45	-	-	V
V _{OL}	Output low-level voltage	-	-	0.45	V
V _{IH}	Input high-level voltage	0.65 * V _{CCQ} ⁽¹⁾	-	V _{CCQ} + 0.3	V
V _{IL}	Input low-level voltage	V _{SS} - 0.3	-	0.35 * V _{CCQ} ⁽²⁾	V

Symbol	Parameter	Min	Typ	Max	Unit
	(1).0.7 * V _{CCQ} for MMC4.3 or lower.				
	(2).0.3 * V _{CCQ} for MMC4.3 or lower.				

4.6 GPADC Electrical Characteristics

Table 4-6 lists the GPADC electrical characteristics.

Table 4-6 GPADC Electrical Characteristics

Parameter	Min	Typ	Max	Unit
ADC Resolution	-	12	-	bits
Full-scale Input Range	0	-	AVCC	V
Sampling Rate	-	-	1	MHz

4.7 TPADC Electrical Characteristics

Table 4-7 lists the TPADC electrical characteristics.

Table 4-7 TPADC Electrical Characteristics

Parameter	Min	Typ	Max	Unit
ADC Resolution	-	12	-	bits
Full-scale Input Range	0	-	AVCC	V
Sampling Rate	-	-	750	kHz

4.8 Audio Codec Electrical Characteristics

Test Conditions

AVCC = 1.8 V, Ta = 25 °C, 1 kHz sinusoid signal, DAC fs = 48 kHz, 16-bit audio data unless otherwise stated.

Table 4-8 Audio Codec Typical Performance Parameters

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
DAC Path	DAC to HPOUTL or HPOUTR(R=100K)					
	Full-scale	0dBFS 1kHz	-	545	-	mVrms
	SNR(A-weighted)	0data	-	99	-	dB
	THD+N	0dBFS 1kHz	-	-85	-	dB
	Crosstalk	R_0dB_L_0data 1kHz L_0dB_R_0data 1kHz	-	-90	-	dB

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
	DAC to HPOUTL or HPOUTR(R=16Ω)					
	Full-scale	0dBFS 1kHz	-	500	-	mVrms
	SNR(A-weighted)	0data	-	99	-	dB
	THD+N(10mW)	1kHz	-	-85	-	dB
	Crosstalk	R_0dB_L_0data 1kHz L_0dB_R_0data 1kHz	-	-90	-	dB
	DAC to HPOUTL or HPOUTR(R=32Ω)					
	Full-scale	0dBFS 1kHz	-	530	-	mVrms
	SNR(A-weighted)	0data	-	99	-	dB
	THD+N(8.8mW)	0dBFS 1kHz	-	-85	-	dB
	Crosstalk	R_0dB_L_0data 1kHz L_0dB_R_0data 1kHz	-	-90	-	dB
	DAC to LINEOUTLP/N or LINEOUTRP/N(R=100K)					
	Full-scale	0dBFS 1kHz	-	1080	-	mVrms
	SNR(A-weighted)	0data	-	100	-	dB
	THD+N	0dBFS 1kHz	-	-90	-	dB
	Crosstalk	R_0dB_L_0data 1kHz L_0dB_R_0data 1kHz	-	-110	-	dB
ADC Path	MICIN via ADC					
	Output Level	MICP=3.3Vpp/2	-	920	-	mFFS
	SNR(A-weighted)	MICN=3.3Vpp/2	-	95	-	dB
	THD+N	1kHz, 0dB Gain	-	-90	-	dB
	Output Level	MICP=1.65Vpp/2	-	920	-	mFFS
	SNR(A-weighted)	MICN=1.65Vpp/2	-	90	-	dB
	THD+N	1kHz, 6dB Gain	-	-90	-	dB
	Output Level	MICP=0.828Vpp/2	-	920	-	mFFS
	SNR(A-weighted)	MICN=0.828Vpp/2	-	80	-	dB
	THD+N	1kHz, 12dB Gain	-	-90	-	dB
	Output Level	MICP=0.414Vpp/2	-	920	-	mFFS
	SNR(A-weighted)	MICN=0.414Vpp/2	-	80	-	dB
	THD+N	1kHz, 18dB Gain	-	-85	-	dB

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
	Output Level	MICP=0.208Vpp/2	-	920	-	mFFS
	SNR(A-weighted)	MICN=0.208Vpp/2	-	85	-	dB
	THD+N	1kHz, 24dB Gain	-	-80	-	dB
	Output Level	MICP=0.104Vpp/2	-	920	-	mFFS
	SNR(A-weighted)	MICN=0.104Vpp/2	-	80	-	dB
	THD+N	1kHz, 30dB Gain	-	-75	-	dB
	Output Level	MICP=0.053Vpp/2	-	920	-	mFFS
	SNR(A-weighted)	MICN=0.053Vpp/2	-	80	-	dB
	THD+N	1kHz, 36dB Gain	-	-70	-	dB
LINEINLR via ADC						
	Output Level	LINEINL=1.65Vpp	-	905	-	mFFS
	SNR(A-weighted)	LINEINR=1.65Vpp	-	95	-	dB
	THD+N	1kHz, 0dB Gain	-	-85	-	dB
FMINLR via ADC						
	Output Level	FMINL=1.65Vpp	-	905	-	mFFS
	SNR(A-weighted)	FMINR=1.65Vpp	-	95	-	dB
	THD+N	1kHz, 0dB Gain	-	-85	-	dB

4.9 MIPI DPHY Electrical Characteristics

This section describes some test parameters for MIPI DPHY based on D-PHY Physical Layer Conformance Test Suite Version 1.00.

Table 4-9 MIPI DPHY CTS Reference

Symbol	Parameters	Min	Typ	Max	Unit
Data Lane HS-TX Parameters					
$V_{OD(0)}$	Data Lane HS-TX Differential Voltages	-	222.4		mV
$V_{OD(1)}$	Data Lane HS-TX Differential Voltages	-	220.6	-	mV
ΔV_{OD}	Data Lane HS-TX Differential Voltage Mismatch	-	3	-	mV
t_R	Data Lane HS-TX 20%-80% Rise Time	-	0.305	-	UI
t_F	Data Lane HS-TX 80%-20% Fall Time		0.301	-	UI

Symbol	Parameters	Min	Typ	Max	Unit
Clock Lane HS-TX Parameters					
$V_{OD(0)}$	Clock Lane HS-TX Differential Voltages	-	264		mV
$V_{OD(1)}$	Clock Lane HS-TX Differential Voltages	-	224	-	mV
ΔV_{OD}	Clock Lane HS-TX Differential Voltage Mismatch	-	40	-	mV
t_R	Clock Lane HS-TX 20%-80% Rise Time	-	0.32	-	UI
t_F	Clock Lane HS-TX 80%-20% Fall Time		0.33	-	UI

4.10 External Clock Source Electrical Characteristics

4.10.1 High-speed Crystal/Ceramic Resonator Characteristics

The high-speed external clock can be supplied with a 24 MHz crystal resonator (oscillation mode). The 24 MHz crystal resonator provides 24 MHz reference clock which is connected to the DXIN and DXOUT terminals.

Table 4-10 High-speed 24 MHz Crystal Requirements

Symbol	Parameter	Min	Typ	Max	Unit
f_{X24M_IN}	Crystal parallel resonance frequency	-	24	-	MHz
	Crystal frequency stability and tolerance at 25 °C ⁽¹⁾	-50	-	+50	ppm
	Oscillation mode	Fundamental			
C_0	Shunt capacitance ⁽²⁾	-	6.5	-	pF

- The 50 ppm frequency stability and tolerance can meet the requirement of the device. We recommend selecting 20 ppm crystal devices. If the REFCLK (24 MHz fanout) is used for Wi-Fi chip, the crystal uses the recommended specification or the specified model for Wi-Fi chip.
- The 6.5 pF is only a simulation value. The crystal shunt capacitance (C_0) is given by the crystal manufacturer.

Table 4-11 Crystal Circuit Parameters

Symbol	Parameter
C_1	C_1 capacitance
C_2	C_2 capacitance
C_L	Equivalent load capacitance, specified by the crystal manufacturer
C_0	Crystal shunt capacitance, specified by the crystal manufacturer

Symbol	Parameter
C _{shunt}	Total shunt capacitance

Frequency stability mainly requires that the total load capacitance (C_L) be constant. The crystal manufacturer typically specifies a total load capacitance which is the series combination of C_1 , C_2 , and C_{shunt} .

The total load capacitance is $C_L = [(C_1 * C_2) / (C_1 + C_2)] + C_{shunt}$.

- C_1 and C_2 represent the total capacitance of the respective PCB trace, load capacitor, and other components (excluding the crystal) connected to each crystal terminal. C_1 and C_2 are usually the same size.
- C_{shunt} is the crystal shunt capacitance (C_0) plus any mutual capacitance ($C_{pkg} + C_{PCB}$) seen across the DXIN and DXOUT signals.

In the application, the crystal resonator and the load capacitors must be placed close to the oscillator pins in order to minimize output distortion and the startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics.

4.11 Interface Timing Characteristics

4.11.1 SMHC Interface Timing

4.11.1.1 HS-SDR Mode



IO voltage is 1.8V or 3.3V.

Figure 4-2 SMHC HS-SDR Mode Output Timing Diagram

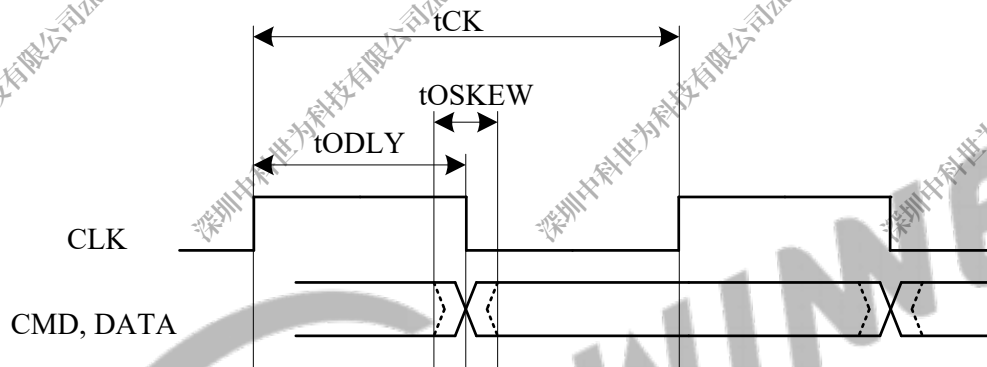


Table 4-12 SMHC HS-SDR Mode Output Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit
CLK					
Clock frequency	tCK	0	50	50	MHz
Duty Cycle	DC	45	50	55	%
Output CMD, DATA(referenced to CLK)					
CMD, Data output delay time	tODLY	-	0.25	0.5	UI
Data output delay skew time	tOSKEW	-	-	0.884	ns
(1) The Unit Interval (UI) is 1-bit nominal time. For example, UI=20 ns at 50 MHz.					
(2) The driver strength level of GPIO is 2 for test.					

Figure 4-3 SMHC HS-SDR Mode Input Timing Diagram

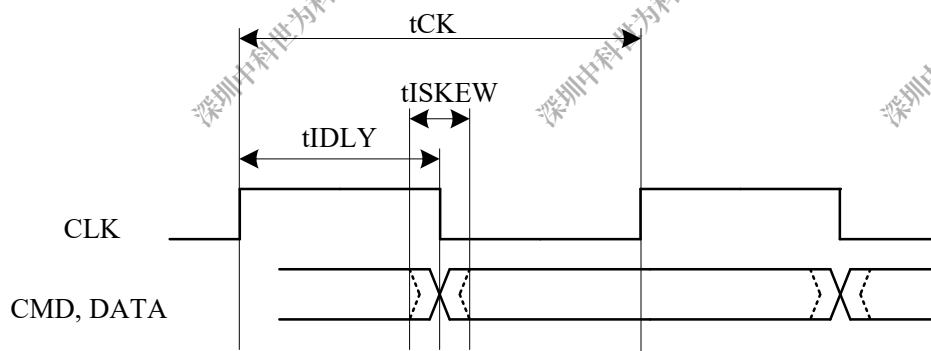


Table 4-13 SMHC HS-SDR Mode Input Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit
CLK					
Clock frequency	tCK	0	50	50	MHz
Duty Cycle	DC	45	50	55	%
Input CMD, DATA(referenced to CLK 50MHz)					
Data input delay in SDR mode. It includes the PCB delay time of Clock, the PCB delay time of Data and the data output delay of Device	tIDLY	-	-	20	ns
Data input skew time in SDR mode	tISKEW	-	-	0.858	ns
The driver strength level of GPIO is 2 for test.					

4.11.1.2 HS-DDR Mode

Figure 4-4 SMHC HS-DDR Mode Output Timing Diagram

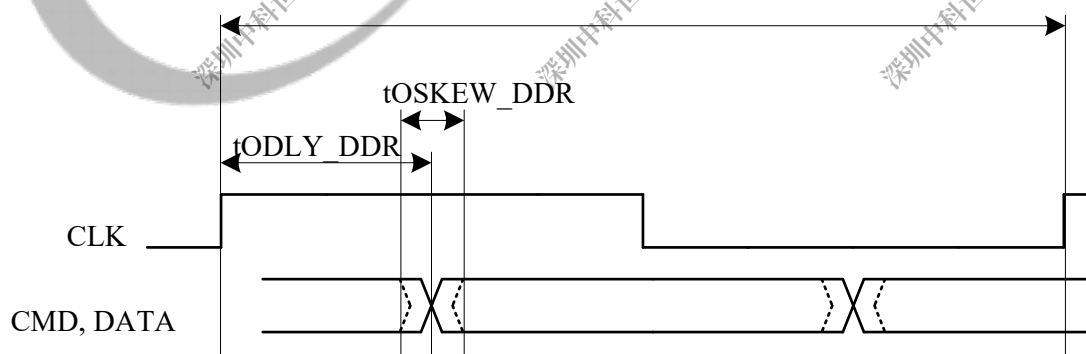


Table 4-14 SMHC HS-DDR Mode Output Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit
CLK					
Clock frequency	tCK	0	50	50	MHz
Duty Cycle	DC	45	50	55	%
Output CMD, DATA(referenced to CLK)					
CMD, Data output delay time in DDR mode	tODLY-DDR	-	0.25	0.25	UI
Data output delay skew time	tOSKEW	-	-	0.884	ns
(1) The Unit Interval (UI) is 1-bit nominal time. For example, UI=20 ns at 50 MHz. (2) The driver strength level of GPIO is 2 for test.					

Figure 4-5 SMHC HS-DDR Mode Input Timing Diagram

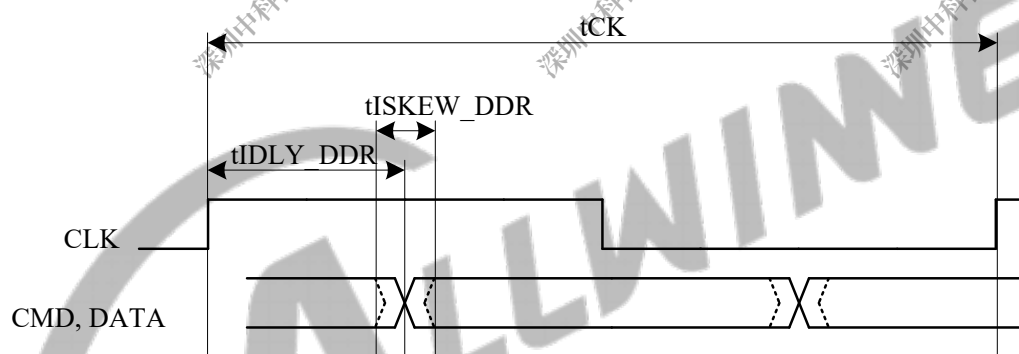


Table 4-15 SMHC HS-DDR Mode Input Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit
CLK					
Clock frequency	tCK	0	50	50	MHz
Duty Cycle	DC	45	50	55	%
Input CMD, DATA(referenced to CLK 50MHz)					
Data input delay in DDR mode. It includes the PCB delay time of Clock, the PCB delay time of Data and the data output delay of Device	tIDLY-DDR	-	-	8.3	ns
Data input skew time in DDR mode	tISKEW-DDR	-	-	0.858	ns
The driver strength level of GPIO is 2 for test.					

4.11.1.3 SDR104 Mode

Figure 4-6 SMHC SDR104 Mode Host Output and Device Input Timing Diagram

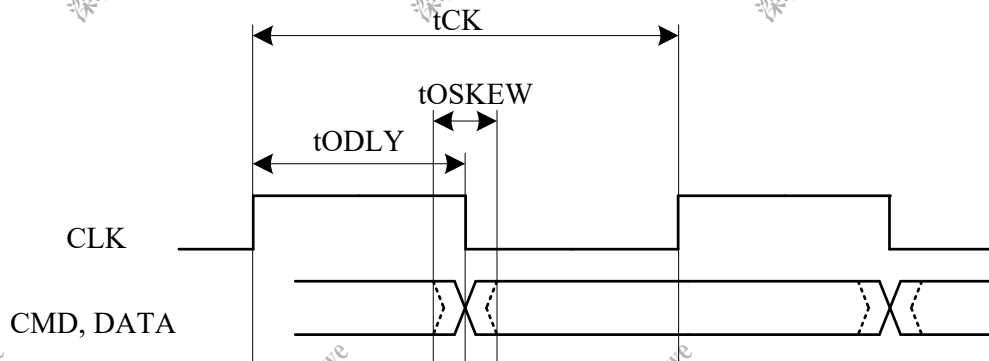


Table 4-16 SMHC SDR104 Mode Host Output and Device Input Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit
CLK					
Clock frequency	tCK	-	-	150	MHz
Duty Cycle	DC	45	50	55	%
Rise time, fall time	tTLH, tTHL	-	-	0.2	UI
Host Output CMD, DATA (referenced to CLK)					
Host CMD, Data output delay time	tODLY	-	0.25	0.5	UI
Host Data output delay skew time	tOSKEW	-	-	0.884	ns
(1) The Unit Interval (UI) is 1-bit nominal time. For example, UI=10 ns at 100 MHz. (2) The driver strength level of GPIO is 3 for test.					

Figure 4-7 SMHC SDR104 Mode Host Input Timing and Device Output Timing Diagram

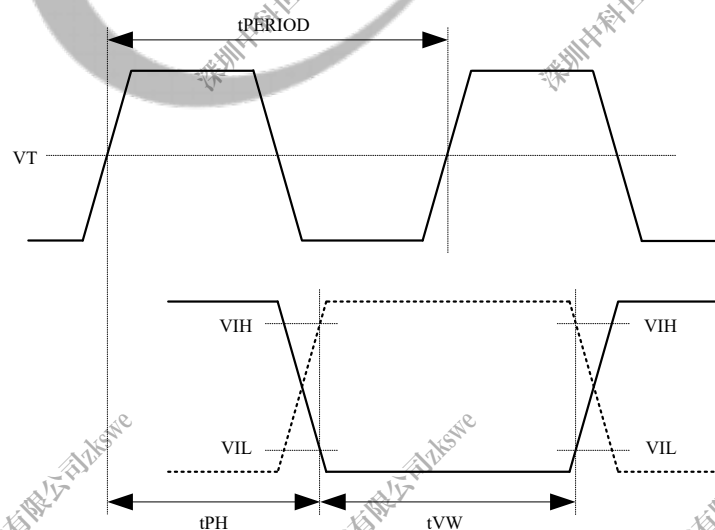


Table 4-17 SMHC SDR104 Mode Host Input Timing and Device Output Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit	Remark
CLK						
Clock Period	tPERIOD	6.66	-	-	ns	Max: 150 MHz
Duty Cycle	DC	45	50	55	%	
Rise time, fall time	tTLH, tTHL	-	-	0.2	UI	
Host Input CMD, DATA (referenced to CLK)						
Device Output delay	tPH	0	-	2	UI	
Device Output delay variation due to temperature change after tuning	dPH	-350 ⁽³⁾	-	1550 ⁽⁴⁾	ps	
Device CMD, Data valid window	tVW	0.575	-	-	UI	
(1) The Unit Interval (UI) is 1-bit nominal time. For example, UI=10 ns at 100 MHz. (2) The driver strength level of GPIO is 3 for test. (3) Temperature variation: -20 °C. (4) Temperature variation: 90 °C.						

4.11.2 LCD Interface Timing

Figure 4-8 HV_IF Vertical Timing

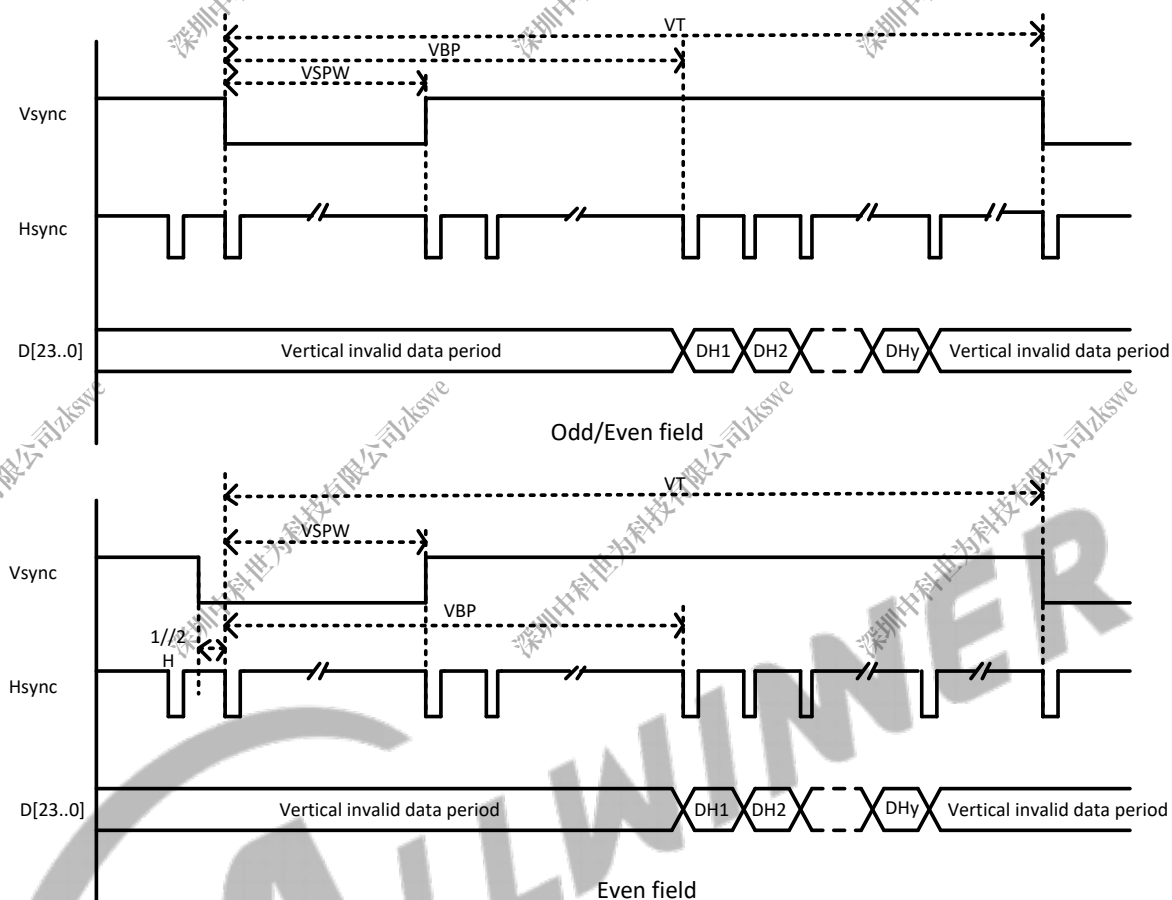


Figure 4-9 HV_IF Horizontal Timing

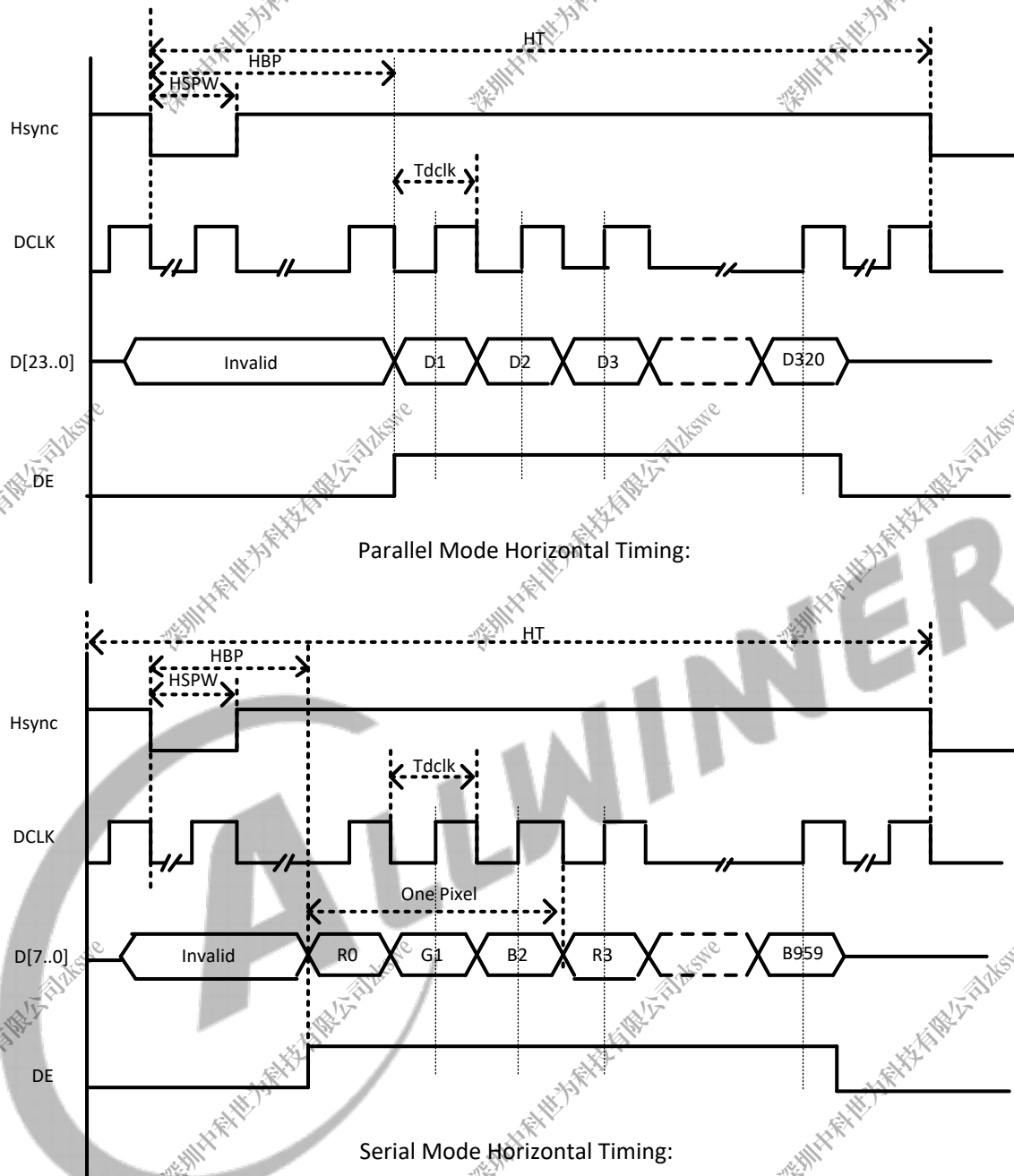


Table 4-18 LCD HV_IF Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit
DCLK Cycle Time	tDCLK	5.9	-	-	ns
Hsync Period Time	tHT	-	HT+1	-	tDCLK
Hsync Width	tHSPW	-	HSPW+1	-	tDCLK
Hsync Back Porch	tHBP	-	HBP+1	-	tDCLK
Vsync Period Time	tVT	-	VT/2	-	tHT
Vsync Width	tVSPW	-	VSPW+1	-	tHT

Parameter	Symbol	Min	Typ	Max	Unit
Vsync Back Porch	tVBP	-	VBP+1	-	tHT
(1) Vsync: Vertical sync, indicates one new frame. (2) Hsync: Horizontal sync, indicate one new scan line. (3) DCLK: Dot clock, pixel data are sync by this clock. (4) LDE: LCD data enable. (5) LD[23..0]: 24Bit RGB/YUV output from input FIFO for panel.					

4.11.3 Parallel CSI Interface Timing

Figure 4-10 Parallel CSI Data Sample Timing

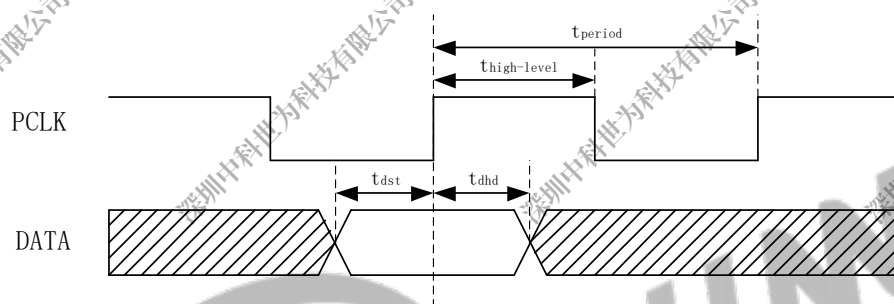


Table 4-19 Parallel CSI Interface Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit
Pclk period	t_{period}	6.73	-	-	ns
Pclk frequency	$1/t_{period}$	-	-	148.5	MHz
Pclk duty	$t_{high-level}/t_{period}$	40	50	60	%
Data input setup time	t_{dst}	0.6	-	-	ns
Data input hold time	t_{dhd}	0.6	-	-	ns

4.11.4 MIPI DPHY Interface Timing

Figure 4-11 MIPI DPHY Timing

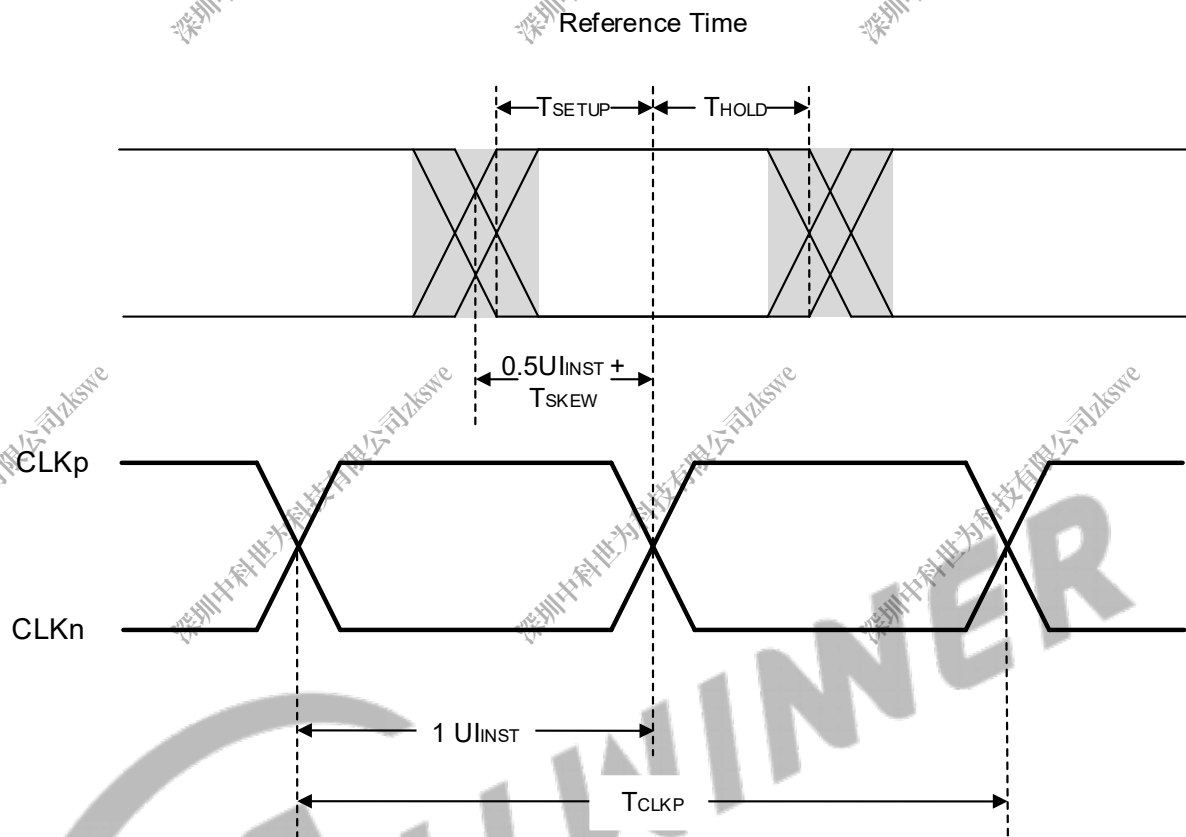


Table 4-20 MIPI DPHY Timing Constants

Parameter	Symbol	Units in U_{INST}			Operational Frequency in Gbps	
		Min	Max	Total	Min	Max
Data to Clock Skew	$T_{skew[tx]}$	-0.15	0.15	0.3	0.08	1.0
		-0.20	0.20	0.4	>1.0	1.5
Data to Clock Setup Time	$T_{setup[rx]}$	0.15	-	-	0.08	1.0
		0.20			>1.0	1.5
Clock to Data Hold Time	$T_{hold[rx]}$	0.15	-	-	0.08	1.0
		0.20			>1.0	1.5

4.11.5 GMAC Interface Timing

4.11.5.1 RGMII

Figure 4-12 RGMII Receive Timing

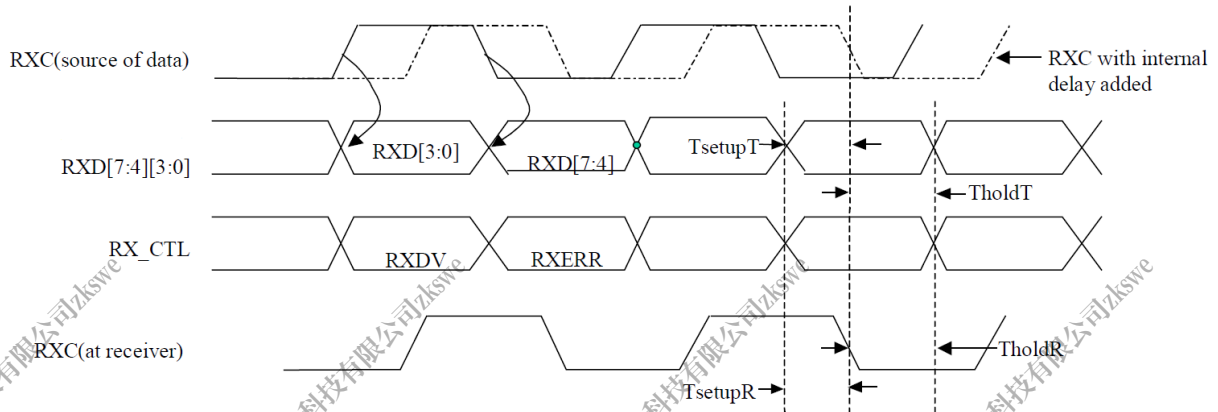


Figure 4-13 RGMII Transmit Timing

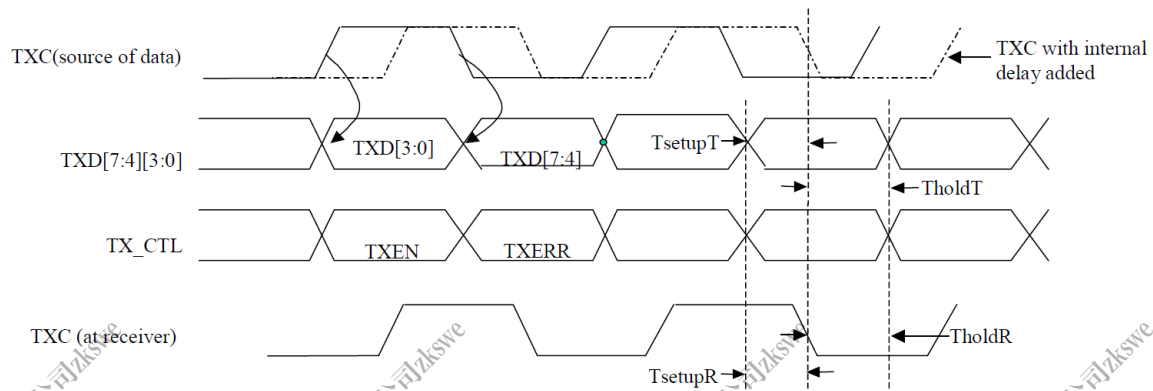


Table 4-21 RGMII Timing Parameters

Parameter	Description	Min	Typical	Max	Unit
Tcyc	Clock Cycle Duration *note1	7.2	8	8.8	ns
Duty_G	Duty Cycle Duration for Gigabit	45	50	55	%
Duty_T	Duty Cycle for 10/100T	40	50	60	%
TsetupT	Data to clock output setup(at Transmitter integrated delay)	1.2	2.0		ns
TholdT	Data to clock output hold(at Transmitter integrated delay)	1.2	2.0		ns
TsetupR	Data to clock input setup(at Receiver integrated delay)	1.0	2.0		ns

Parameter	Description	Min	Typical	Max	Unit
TholdR	Data to clock input hold(at Receiver integrated delay)	1.0	2.0		ns
For 10Mbps and 100Mbps, T _{cy} will scale 400ns±40ns and 40ns±4ns.					

4.11.5.2 RMII

Figure 4-14 RMII Receive Timing

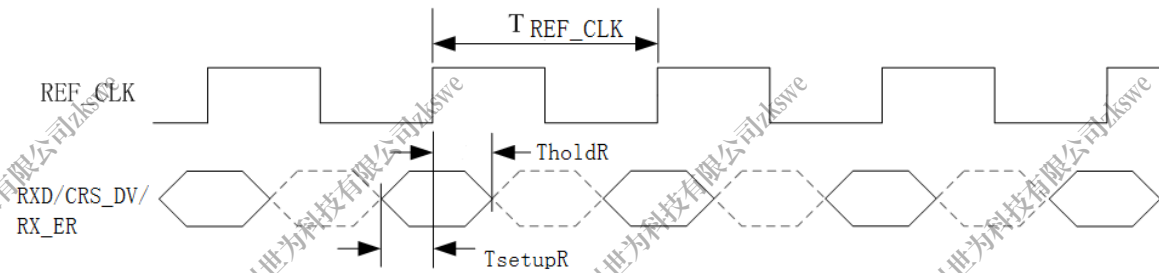


Figure 4-15 RMII Transmit Timing

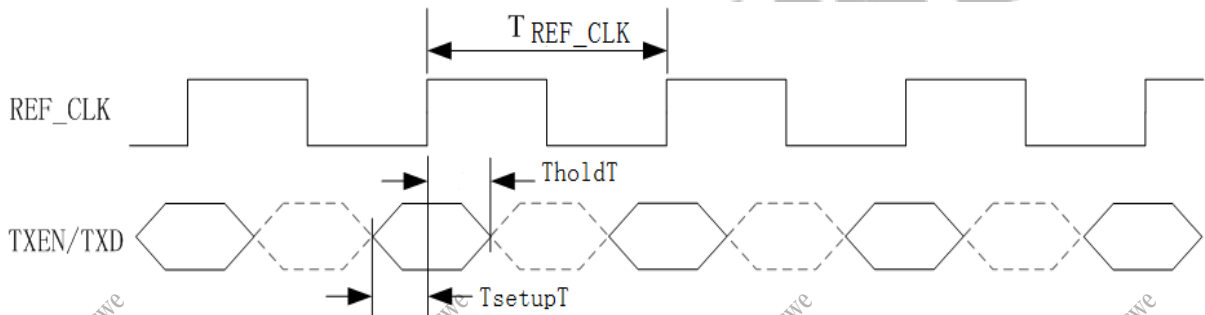


Table 4-22 RMII Timing Parameters

Parameter	Description	Min	Typ	Max	Unit
T _{REF_CLK}	Reference Clock Period	-	20	-	ns
T _{duty}	REF_CLK duty cycle	35		65	%
T _{setupT}	TXD/TXEN to REF_CLK setup time	4			ns
TholdT	TXD/TXEN to REF_CLK hold time	2			ns
T _{setupR}	RXD/CRS_DV/RX_ER to REF_CLK setup time	4			ns
TholdR	RXD/CRS_DV/RX_ER to REF_CLK hold time	2			ns

4.11.6 SPI Interface Timing

Figure 4-16 SPI Writing Timing

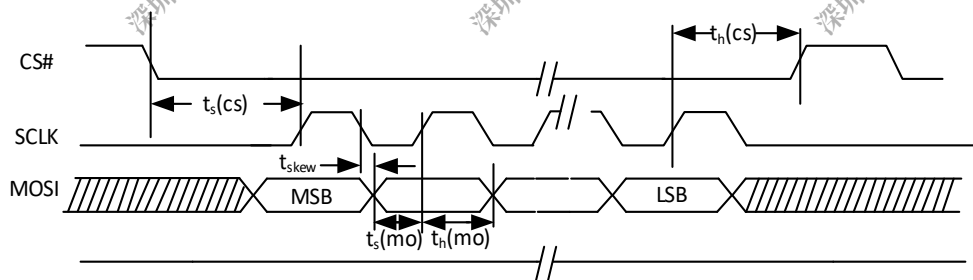


Figure 4-17 SPI Reading Timing

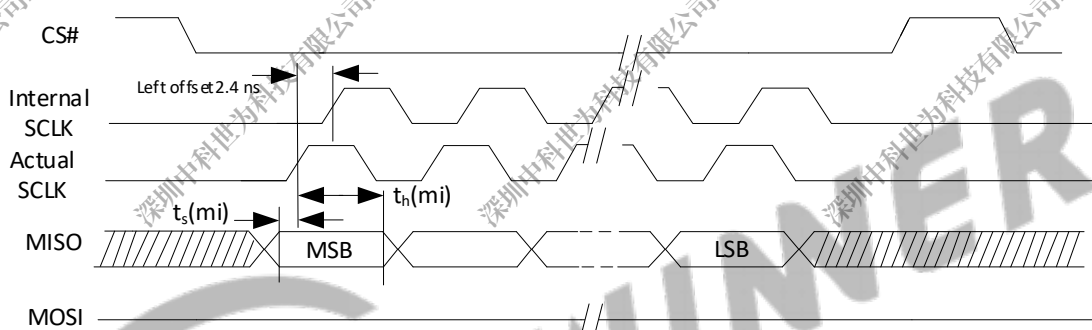


Table 4-23 SPI Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit
CS# Active Setup Time	$t_s(cs)$	$2T^{(1)}$	-	-	ns
CS# Active Hold Time	$t_h(cs)$	$2T^{(1)}$	-	-	ns
CLK/Data output skew time	$t_{skew}^{(2)}$	-	-	1	ns
Data output Delay Time	$t_s(mo)$	$T^{(1)}/2 - t_{skew}$	-	$T^{(1)}/2 + t_{skew}$	ns
Data output Hold Time	$t_h(mo)$	$T^{(1)}/2 - t_{skew}$	-	$T^{(1)}/2 + t_{skew}$	ns
Data In Setup Time	$t_s(mi)$	0.2(master mode) 4.5(slave mode)	-	-	ns
Data In Hold Time	$t_h(mi)$	0.2(master mode) 2(slave mode)	-	-	ns

(1) T is the cycle of clock.

(2) t_{skew} is the output skew time between SCLK and MOSI. SCLK may be either faster or slower than MOSI.

4.11.7 SPI-DBI Interface Timing

Figure 4-18 DBI 3-line Serial Interface Timing

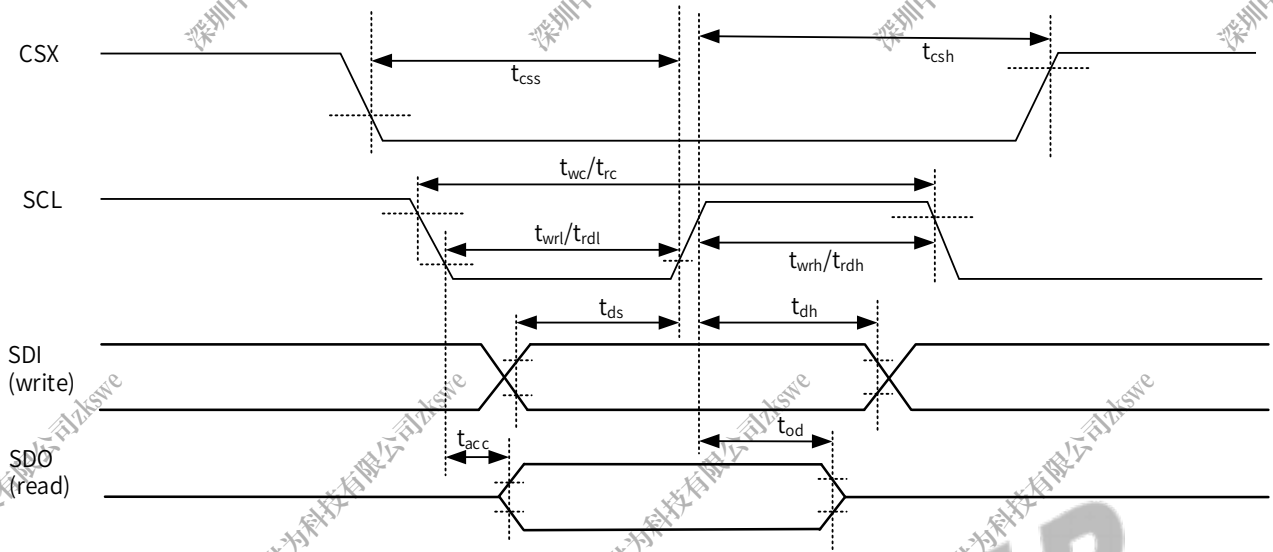


Table 4-24 DBI 3-line Serial Interface Write Timing Parameters

Signal	Parameter	Symbol	Min	Max	Unit
CSX	Chip select setup time	t_{css}	15		ns
SCL	Write cycle	t_{wc}	16		ns
	Control pulse “H” duration	t_{wrh}	7		ns
	Control pulse “L” duration	t_{wrl}	7		ns
SDI/SDO	Data setup time	t_{ds}	7 ⁽¹⁾		ns
	Data hold time	t_{dt}	7 ⁽¹⁾		ns

Note:

Range of required clock frequency: 0-60 MHz.

Table 4-25 DBI 3-line Serial Interface Read Timing Parameters

Signal	Parameter	Symbol	Min	Max	Unit
CSX	Chip select setup time	t_{csh}	60		ns
SCL	Read cycle	t_{rc}	150		ns
	Control pulse “H” duration	t_{rdh}	60		ns
	Control pulse “L” duration	t_{rdl}	60		ns
SDI/SDO	Read access time	t_{racc}	10 ⁽¹⁾	50	ns
	Output disable time	t_{od}	15 ⁽¹⁾	50	ns

Signal	Parameter	Symbol	Min	Max	Unit
Note: Range of required clock frequency: 0-6.67 MHz.					

Figure 4-19 DBI 4-line Serial Interface Timing

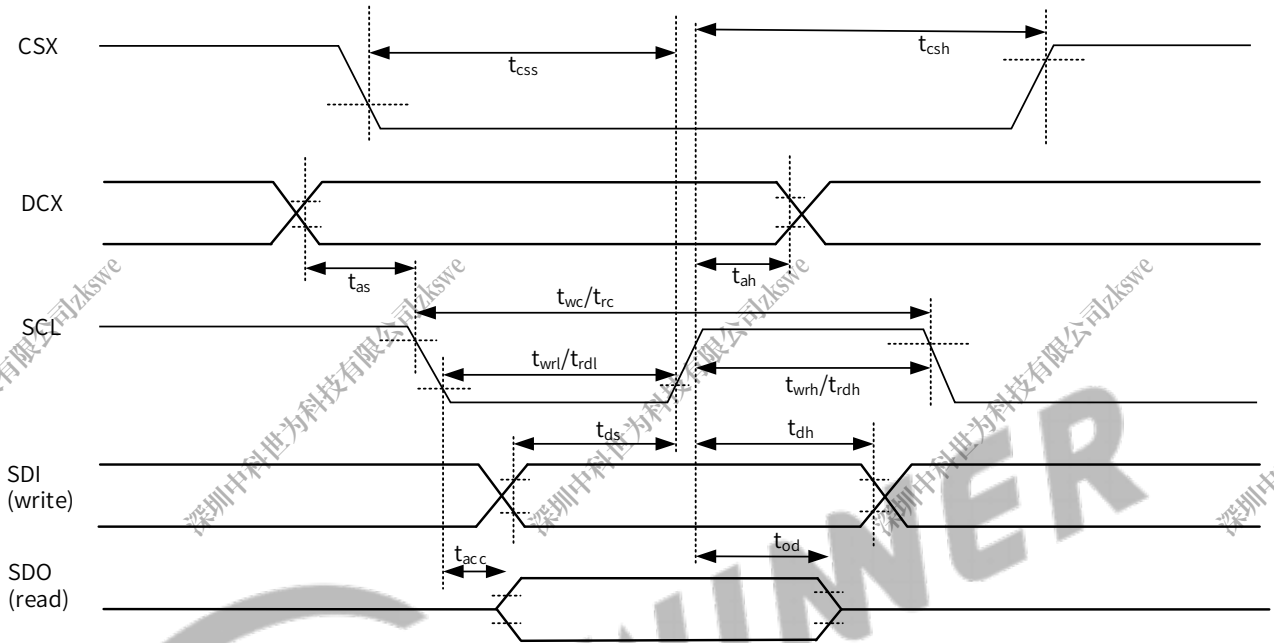


Table 4-26 DBI 4-line Serial Interface Write Timing Parameters

Signal	Parameter	Symbol	Min	Max	Unit
CSX	Chip select setup time	t_{css}	15		ns
DCX	Address setup time	t_{as}	10		ns
	Address hold time	t_{ah}	10		ns
SCL	Write cycle	t_{wc}	16		ns
	Control pulse “H” duration	t_{wrh}	7		ns
	Control pulse “L” duration	t_{wrl}	7		ns
SDI/SDO	Data setup time	t_{ds}	7 ⁽¹⁾		ns
	Data hold time	t_{dt}	7 ⁽¹⁾		ns
	Output disable time	t_{od}	15 ⁽¹⁾	50	ns
Note: Range of required clock frequency: 0-60 MHz.					

Table 4-27 DBI 4-line Serial Interface Read Timing Parameters

Signal	Parameter	Symbol	Min	Max	Unit
CSX	Chip select setup time	t_{csh}	60		ns

Signal	Parameter	Symbol	Min	Max	Unit
DCX	Address setup time	t_{as}	10		ns
	Address hold time	t_{ah}	10		ns
SCL	Read cycle	t_{rc}	150		ns
	Control pulse “H” duration	t_{rdh}	60		ns
	Control pulse “L” duration	t_{rdl}	60		ns
SDI/SDO	Read access time	t_{racc}	-	50	ns
	Output disable time	t_{od}	15 ⁽¹⁾	50	ns
Note: Range of required clock frequency: 0-6.67 MHz.					

4.11.8 UART Interface Timing

Figure 4-20 UART RX Timing

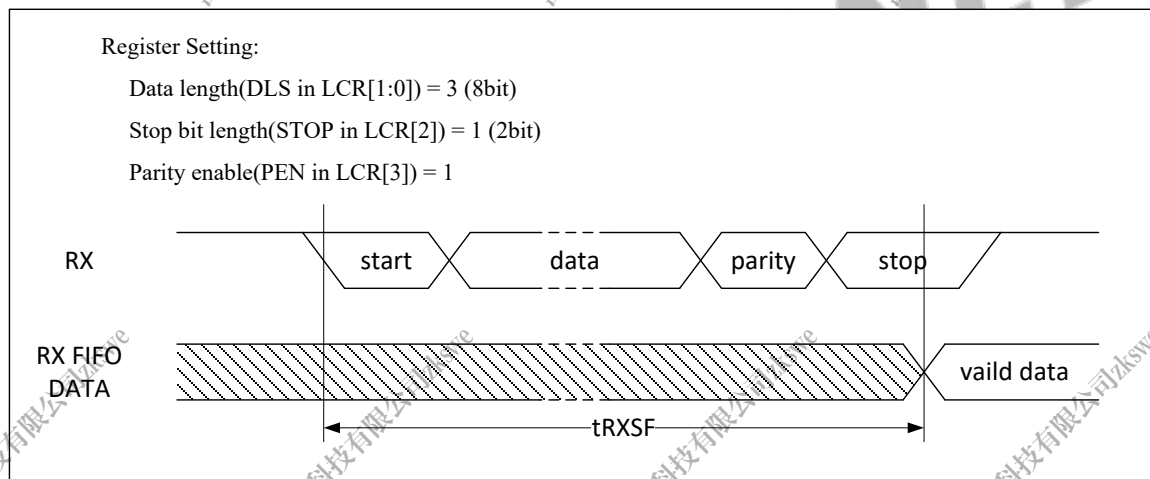


Figure 4-21 UART nCTS Timing

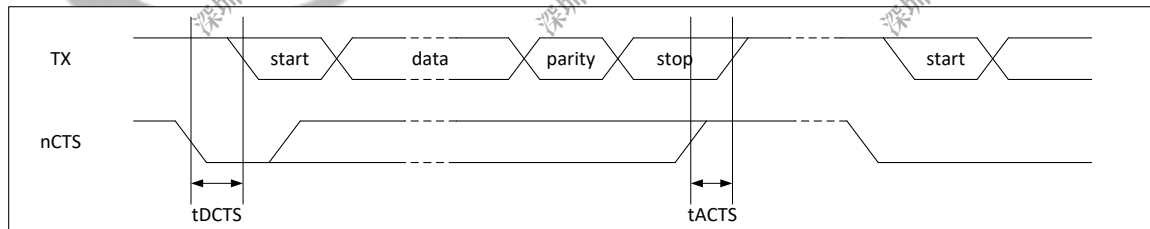


Figure 4-22 UART nRTS Timing

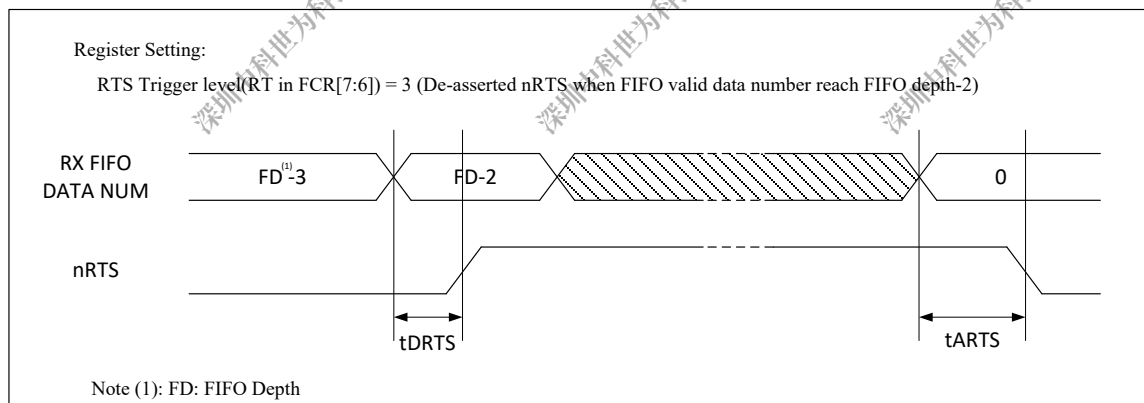


Table 4-28 UART Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit
RX start to RX FIFO	tRXSF	10.5*BRP ⁽¹⁾	-	11*BRP ⁽¹⁾	ns
Delay time of de-asserted nCTS to TX strat	tDCTS	-	-	BRP ⁽¹⁾	ns
Step time of asserted nCTS to stop next transmission	tACTS	BRP ⁽¹⁾ / 4	-	-	ns
Delay time of de-asserted nRTS	tDRTS	-	-	BRP ⁽¹⁾	ns
Delay time of asserted nRTS	tARTS	-	-	BRP ⁽¹⁾	ns

(1) BRP: Baud-Rate Period.

4.11.9 TWI Interface Timing

Figure 4-23 TWI Timing

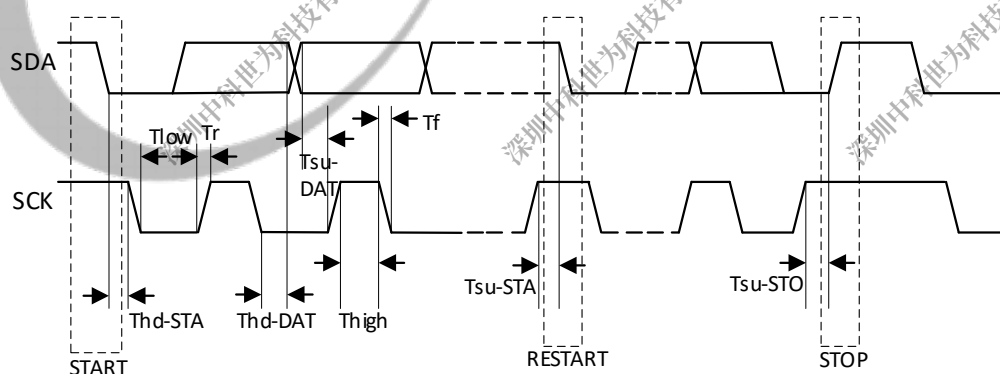


Table 4-29 TWI Timing Parameters

Parameter	Symbol	Standard mode		Fast mode		Unit
		Min	Max	Min	Max	
SCK clock frequency	Fsck	0	100	0	400	kHz

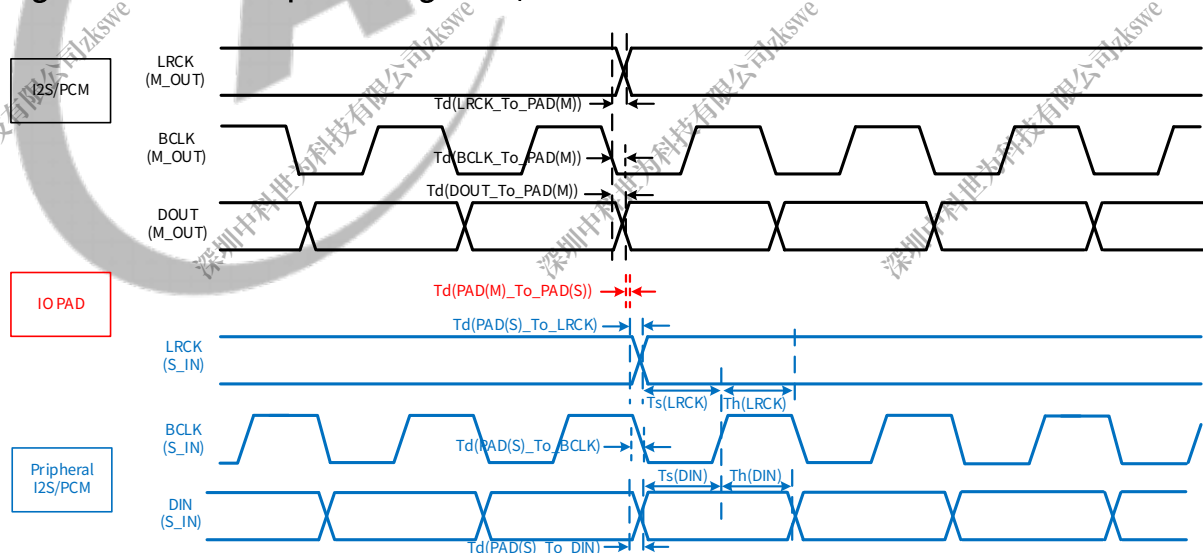
Parameter	Symbol	Standard mode		Fast mode		Unit
		Min	Max	Min	Max	
Setup Time In Start	Tsu-STA	4.7	-	0.6	-	us
Hold Time In Start	Thd-STA	4.0	-	0.6	-	us
Setup Time In Data	Tsu-DAT	250	-	100	-	ns
Hold Time In Data	Thd-DAT	0	-	0	-	us
Setup Time In Stop	Tsu-STO	4.0	-	0.6	-	us
SCK Low level Time	Tlow	4.7	-	1.3	-	us
SCK High level Time	Thigh	4.0	-	0.6	-	us
SCK/SDA Falling Time	Tf	-	300	20	300	ns
SCK/SDA Rising Time	Tr	-	1000	20	300	ns

4.11.10 I2S/PCM Interface Timing

4.11.10.1 Data Output timing of I2S/PCM in Master mode

The Data Output timing of I2S/PCM in Master mode and the Data Input timing of Peripheral I2S/PCM in Slave mode show in Figure 4-24.

Figure 4-24 Data Output Timing of I2S/PCM in Master Mode



The Data Output timing parameters of I2S/PCM in Master mode and The Data Input timing parameters of Peripheral I2S/PCM in Slave mode show in Table 4-30.

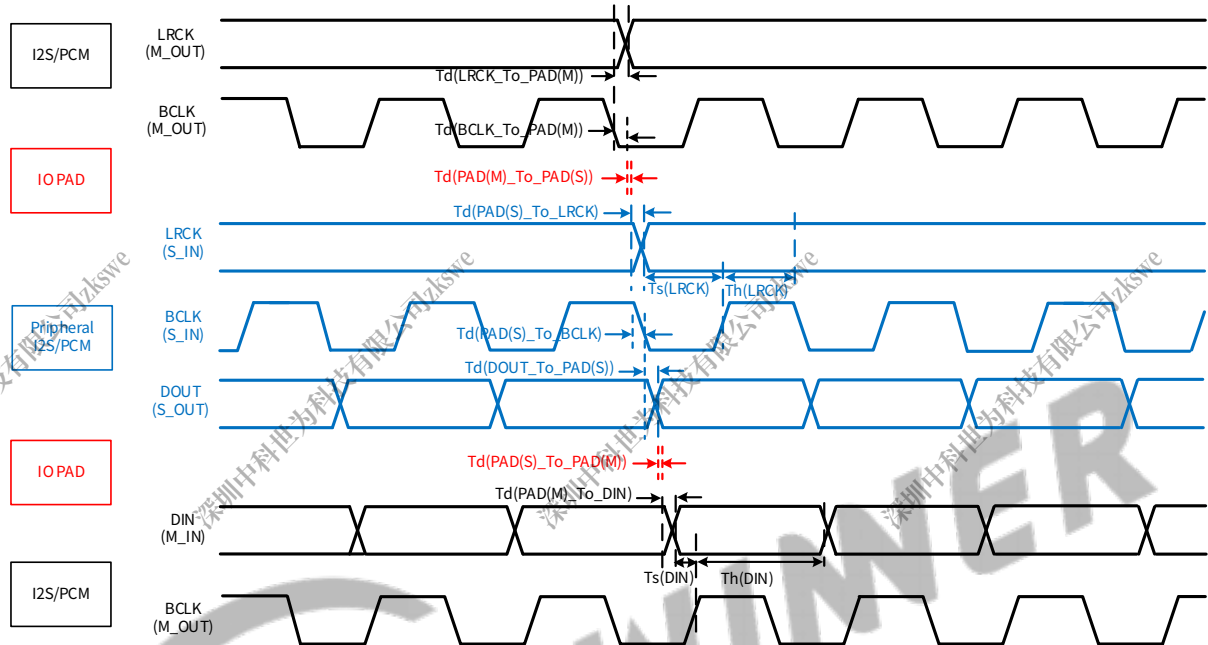
Table 4-30 Data Output Timing Parameters of I2S/PCM in Master Mode

Parameter		Min	Max	Skew	Units
Sequence requirement of internal signal of I2S/PCM in Master mode					
T _d (LRCK_To_PA D(M))	LRCK to PAD(M) Delay	/	T1<6.5 (restriction)	T _w 1<2.5(require ment)	ns
T _d (BCLK_To_PA D(M))	BCLK to PAD(M) Delay	/	T2<6.5(restricti on)		
T _d (DOUT_To_PA D(M))	DOUT to PAD(M) Delay	/	T3<6.5(restricti on)		
Sequence requirement of the IO pad of I2S/PCM in Master mode connecting to the external IO pad of Peripheral I2S/PCM in Slave mode					
T _d (PAD(M)_To_ PAD(S))	LRCK PAD(M) to LRCK PAD(S) Delay	/	T4*<7.0(estima tion)	T _w 2*<1.0(require ment)	ns
	BCLK PAD(M) to BCLK PAD(S) Delay	/	T5*<7.0(estima tion)		
	DOUT PAD(M) to DIN PAD(S) Delay	/	T6*<7.0(estima tion)		
Sequence requirement of internal signal of Peripheral I2S/PCM in Slave mode					
T _d (PAD(S)_To_L RCK)	PAD(S) to LRCK Delay	/	T7*<6.5(assum ption)	T _w 3*<2.5(require ment)	ns
T _d (PAD(S)_To_B CLK)	PAD(S) to BCLK Delay	/	T8*<6.5(assum ption)		
T _d (PAD(S)_To_D IN)	PAD(S) to DIN Delay	/	T9*<6.5(assum ption)	/	/
T _s (LRCK)	LRCK Setup Slack	T10*(anal ysis)	/	/	ns
T _h (LRCK)	LRCK Hold Slack	T11*(anal ysis)	/	/	ns
T _s (DIN)	DIN Setup Slack	T12*(anal ysis)	/	/	ns
T _h (DIN)	DIN Hold Slack	T13*(anal ysis)	/	/	ns

4.11.10.2 Data Input timing of I2S/PCM in Master mode

The Data Input timing of I2S/PCM in Master mode and the Data Output timing of Peripheral I2S/PCM in Slave mode show in Figure 4-25.

Figure 4-25 Data Input Timing of I2S/PCM in Master Mode



The Data Input timing parameters of I2S/PCM in Master mode and The Data Output timing parameters of Peripheral I2S/PCM in Slave mode are shown in Table 4-31.

Table 4-31 Data Input Timing Parameters of I2S/PCM in Master Mode

Parameter		Min	Max	Skew	Units
Sequence requirement of internal signal of I2S/PCM in Master mode					
Td(LRCK_To_PAD(M))	LRCK to PAD(M) Delay	/	T1<6.5(requirement)	Tw1<2.5 (estimation)	ns
Td(BCLK_To_PAD(M))	BCLK to PAD(M) Delay	/	T2<6.5(requirement)		ns
Sequence requirement of the IO pad of I2S/PCM in Master mode connecting to the external IO pad of Peripheral I2S/PCM in Slave mode					
Td(PAD(M)_To_PAD(S))	LRCK PAD(M) to LRCK PAD(S) Delay	/	T3*<7.0(requirement)	Tw2*<1.0 (estimation)	ns
	BCLK PAD(M) to BCLK PAD(S) Delay		T4*<7.0(requirement)		

Parameter		Min	Max	Skew	Units
Sequence requirement of internal signal of Peripheral I2S/PCM in Slave mode					
Td(PAD(S)_To_LRC K)	PAD(S) to LRCK Delay	/	$T5^* < 6.5$ (requiremen t)	$Tw3^* < 2.5$ (estimati on)	ns
Td(PAD(S)_To_BCL K)	PAD(S) to BCLK Delay	/	$T6^* < 6.5$ (requiremen t)		ns
Td(DOUT_To_PAD(S)	DOUT to PAD(S) Delay	/	$T7^* < 6.5$ (requiremen t)	/	ns
Ts(LRCK)	LRCK Setup Slack	$T8^*$ (analys is)	/	/	ns
Th(LRCK)	LRCK Hold Slack	$T9^*$ (analys is)	/	/	ns
Sequence requirement of the IO pad of I2S/PCM in Master mode connecting to the external IO pad of Peripheral I2S/PCM in Slave mode					
Td(PAD(S)_To_PAD(M))	DOUT PAD(S) to DIN PAD(M) Delay	/	$T10^* < 7.0$ (requireme nt)	/	ns
Sequence requirement of internal signal of I2S/PCM in Master mode					
Td(PAD(M)_To_DIN)	PAD(M) to DIN Delay	/	$T11 < 6.5$ (requiremen t)	/	ns
Ts(DIN)	DIN Setup Slack	$T12^*$ (anal ysis)	/	/	ns
Th(DIN)	DIN Hold Slack	$T13^*$ (anal ysis)	/	/	ns

4.11.11 DMIC Interface Timing

Figure 4-26 DMIC Timing

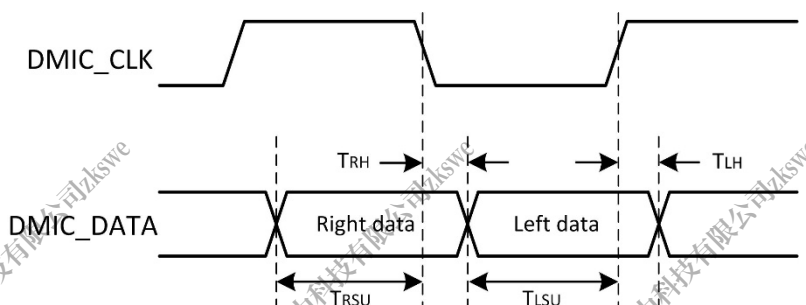


Table 4-32 DMIC Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit
DMIC_DATA(Right) Setup Time to Falling Edge of DMIC_CLK	T_{RSU}	15	-	-	ns
DMIC_DATA(Right) Hold Time from Falling Edge of DMIC_CLK	T_{RH}	0	-	-	ns
DMIC_DATA(Left) Setup Time to Rising Edge of DMIC_CLK	T_{LSU}	15	-	-	ns
DMIC_DATA(Left) Hold Time From Rising edge of DMIC_CLK	T_{LH}	0	-	-	ns

4.11.12 OWA Interface Timing

Figure 4-27 OWA Timing

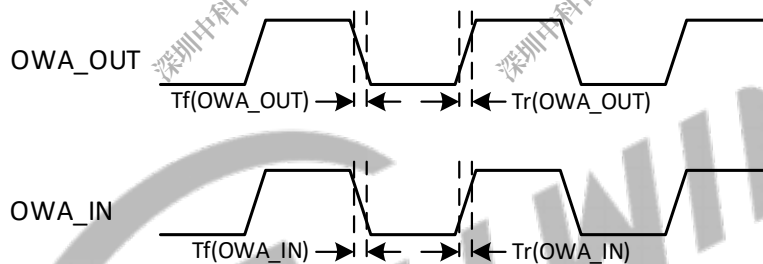


Table 4-33 OWA Timing Constants

Parameter	Symbol	Min	Typ	Max	Unit
OWA_OUT Rise Time	$Tr(OWA_OUT)$	-	-	8	ns
OWA_OUT Fall Time	$Tf(OWA_OUT)$	-	-	8	ns
OWA_IN Rise Time	$Tr(OWA_IN)$	-	-	4	ns
OWA_IN Fall Time	$Tf(OWA_IN)$	-	-	4	ns

4.12 Power-Up and Power-Down Sequence

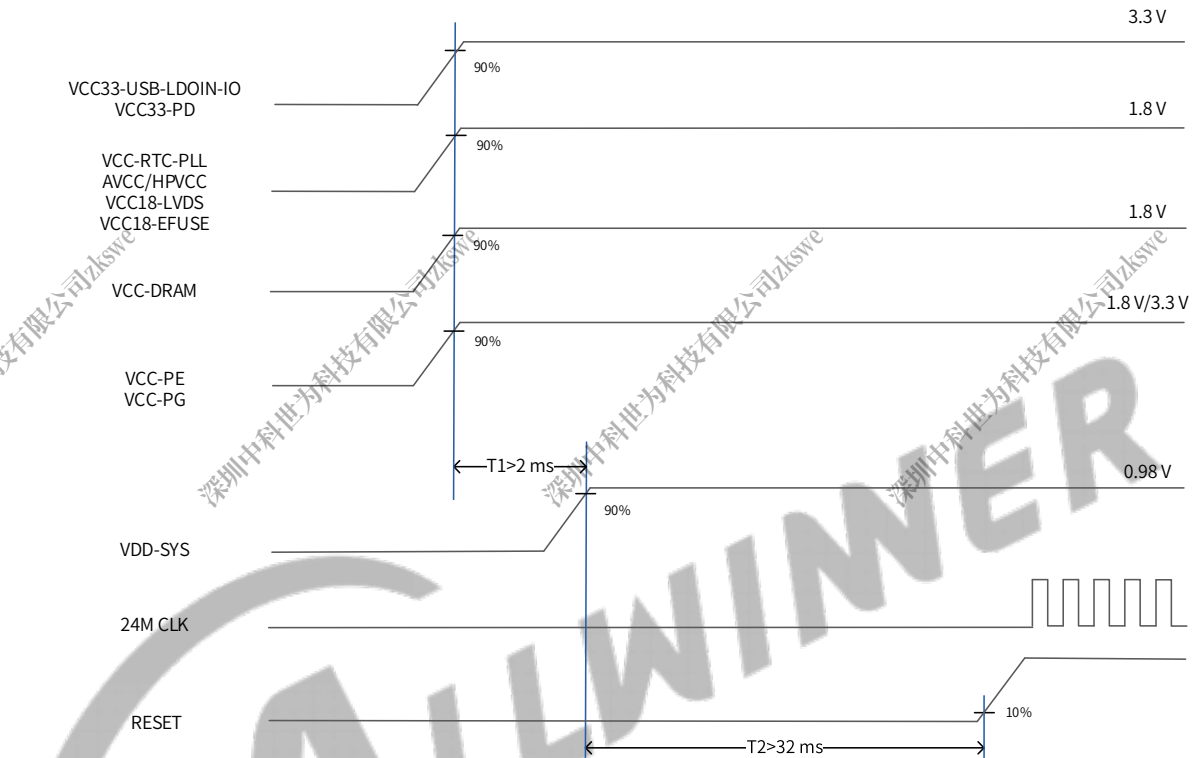
4.12.1 Power-Up Sequence

The following figure shows an example of the power-up sequence for the SoC. The description of the power-up sequence is as follows.

- The consequent steps in power-on sequence should not start before the previous step supplies have been stabilized within 90–110% of their nominal voltage, unless stated otherwise.
- VCC33-USB-LDOIN_IO must be ramped before VDD-SYS with a minimum delay of 2 ms.

- VCC-DRAM needs be stable before SDRAM driver initialization.
- During the entire power on sequence, the RESET signal must be held on low until all other power rails (except 24 MHz CLK) are stable for more than 32 ms.
- 24MHz clock starts oscillating after the RESET signal is released.

Figure 4-28 Power-Up Sequence



NOTE

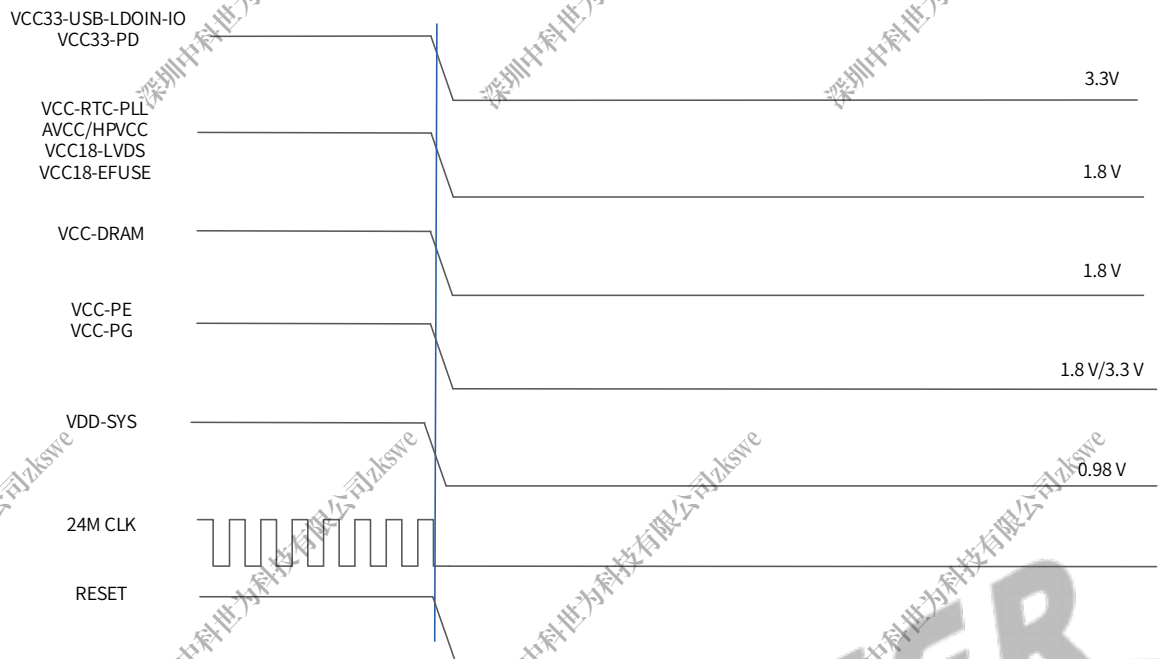
When some of PD0-PD19 I/Os are used as LVDS or DSI, and some are used as GPIO, the power-up sequence of VCC18-LVDS should be advanced to that of VCC33-PD, or the power-up sequence of VCC33-PD should be delayed after that of VCC18-LVDS.

4.12.2 Power-Down Sequence

The power-down requirements are as follows.

- After the RESET signal goes low, the 24 MHz clock starts to stop oscillating.
- No special restrictions for other power rails.

Figure 4-29 Power-Down Sequence



5 Temperature and Thermal Characteristics

5.1.1 Temperature

The following tables describe the temperature of the device

Table 5-1 Operating and Storage Temperature

Symbol	Parameter	Min	Max	Unit
T _a	Ambient Operating Temperature	-25	75	°C
T _{STG}	Storage Temperature	-40	150	°C

Table 5-2 Junction Temperature

Chip	Recommended Operating Temperature (T _J)		Absolute Maximum Junction Temperature	Unit
	Min	Max		
F136	-25	115	125	°C

5.1.2 Package Thermal Characteristics

The maximum chip junction temperature (T_{Jmax}) must never exceed the values given in Table 5-2 Junction Temperature.

Failure to maintain a junction temperature within the range specified reduces operating lifetime, reliability, and performance, and may cause irreversible damage to the system. It is useful to calculate the exact power consumption and junction temperature to determine which the temperature will be best suited to the application. Therefore, the product should include thermal analysis and thermal design to ensure the operating junction temperature of the device is within functional limits.

The following tables show the thermal resistance characteristics of the device. These data are based on JEDEC JESD51 standard, because the actual system design and temperature could be different from JEDEC JESD51, these simulating data are a reference only and may not represent actual use-case values, please prevail in the actual application condition test.

Table 5-3 Package Thermal Characteristics

Symbol	Parameter	Min	Typ ⁽¹⁾⁽²⁾	Max	Unit
θ _{JA}	Junction-to-Ambient Thermal Resistance	-	23.8	-	°C/W
θ _{JB}	Junction-to-Board Thermal Resistance	-	18.44	-	°C/W
θ _{JC}	Junction-to-Case Thermal Resistance	-	4	-	°C/W

(1) Reference document: JESD51-2 Integrated Circuits Thermal Test Method Environment Conditions – Natural Convection (Still Air). Available from www.jedec.org.

- (2) The testing PCB is 4 layers, 114.5 mm x 76.2 mm body, and 1.6 mm thickness. Ambient temperature is 25 °C.

6.1.1 Pin Map

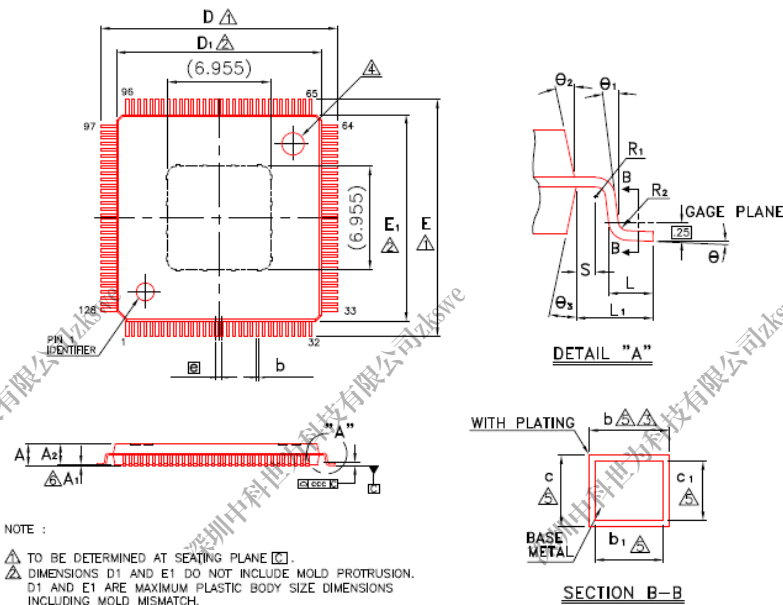
Figure 6-1 Pin Map

[illegible]

6.1.2 Package Dimension

The following figure shows the top, bottom, and side views of the package.

Figure 6-2 Package Dimension



Symbol	Dimension in mm			Dimension in inch		
	Min	Nom	Max	Min	Nom	Max
A	—	—	1.60	—	—	0.063
A1	0.05	—	0.15	0.002	—	0.006
A2	1.35	1.40	1.45	0.053	0.055	0.057
b	0.13	0.18	0.23	0.005	0.007	0.009
b1	0.13	0.16	0.19	0.005	0.006	0.007
c	0.09	—	0.20	0.004	—	0.008
c1	0.09	—	0.16	0.004	—	0.006
D	15.85	16.00	16.15	0.624	0.630	0.636
D1	13.90	14.00	14.10	0.547	0.551	0.555
E	15.85	16.00	16.15	0.624	0.630	0.636
E1	13.90	14.00	14.10	0.547	0.551	0.555
□	0.40	BSC	—	0.016	BSC	—
L	0.45	0.60	0.75	0.018	0.024	0.030
L1	1.00	REF	—	0.039	REF	—
R1	0.08	—	—	0.003	—	—
R2	0.08	—	0.20	0.003	—	0.008
S	0.20	—	—	0.008	—	—
θ	0°	3.5°	7°	0°	3.5°	7°
θ1	0°	—	—	0°	—	—
θ2	11°	12°	13°	11°	12°	13°
θ3	11°	12°	13°	11°	12°	13°
ccc	0.08	—	—	0.003	—	—

TITLE : 128UD E-PAD LFP (14x14x1.4 mm) PACKAGE OUTLINE-Cu L/FFOOTPRINT 2.0mm			
APPR.			
PE.		DWG NO.	PCO2035
PD.		REV NO.	A
QM.		DATE	05/22/'15
CHK.		DWG.	Leo Deng

7 Carrier, Storage and Backing Information

7.1 Carrier

The following table shows matrix tray carrier information.

Table 7-1 Matrix Tray Carrier Information

Item	Color	Size	Note
Tray	Black	322.6 mm x 135.9 mm x 7.62 mm	90 Qty/Tray
Aluminum foil bags	Silvery white	540 mm x 300 mm x 0.14 mm	Vacuum packing Including HIC and desiccant Printing: RoHS symbol
Pearl cotton cushion (Vacuum bag)	White	12 mm x 680 mm x 185 mm	
Pearl cotton cushion (The Gap between vacuum bag and inner box)	White	Left-Right: 12 mm x 180 mm x 85 mm Front-Back: 12 mm x 350 mm x 70 mm	
Inner Box	White	396 mm x 196 mm x 96 mm	Printing: RoHS symbol 10 Tray/Inner box
Carton	White	420 mm x 410 mm x 320 mm	6 Inner box/Carton

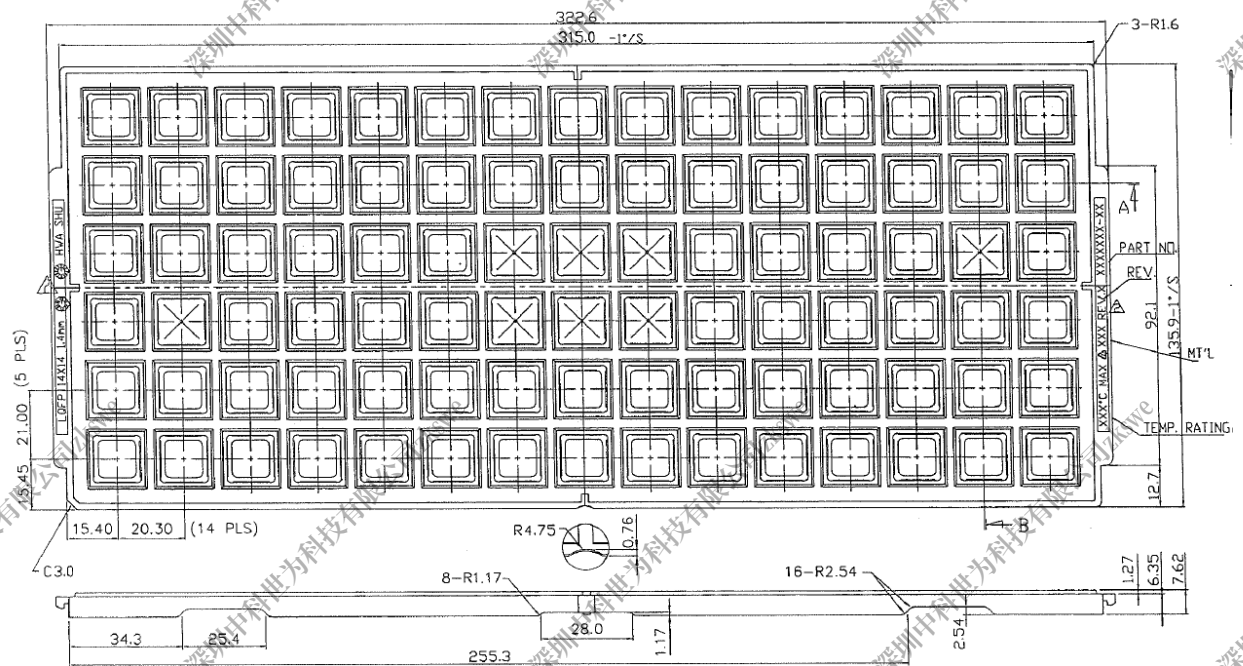
The following table shows packing quantity.

Table 7-2 Packing Quantity Information

Sample	Size	Qty/Tray	Tray/Inner Box	Full Inner Box Qty	Inner Box/Carton	Full Carton Qty
F136	14 mm x 14 mm	90	10	900	6	5400

The following figure shows tray dimension drawing.

Figure 7-1 Tray Dimension Drawing



7.2 Storage

Reliability is affected if any condition specified in section 7.2.2 and section 7.2.3 has been exceeded.

7.2.1 Moisture Sensitivity Level (MSL)

A package's MSL indicates its ability to withstand exposure after it is removed from its shipment bag, a low MSL device sample can be exposed on the factory floor longer than a high MSL device sample. Table 7-3 defines all MSL.

Table 7-3 MSL Summary

MSL	Out-of-bag floor life	Comments
1	Unlimited	$\leq 30^{\circ}\text{C} / 85\%\text{RH}$
2	1 year	$\leq 30^{\circ}\text{C} / 60\%\text{RH}$
2a	4 weeks	$\leq 30^{\circ}\text{C} / 60\%\text{RH}$
3	168 hours	$\leq 30^{\circ}\text{C} / 60\%\text{RH}$
4	72 hours	$\leq 30^{\circ}\text{C} / 60\%\text{RH}$
5	48 hours	$\leq 30^{\circ}\text{C} / 60\%\text{RH}$
5a	24 hours	$\leq 30^{\circ}\text{C} / 60\%\text{RH}$
6	Time on Label (TOL)	$\leq 30^{\circ}\text{C} / 60\%\text{RH}$

**NOTE**

The F136 device samples are classified as MSL3.

7.2.2 Bagged Storage Conditions

The following table defines the shelf life.

Table 7-4 Bagged Storage Conditions

Packing Mode	Vacuum packing
Storage Temperature	20 26°C
Storage Humidity	40% 60%RH
Shelf Life	12 months

7.2.3 Out-of-bag Duration

It is defined by the device MSL rating. The out-of-bag duration is as follows.

Table 7-5 Out-of-bag Duration

Storage Temperature	20 26°C
Storage Humidity	40% 60%RH
Moisture Sensitive Level (MSL)	3
Floor Life	168 hours

For no mention of storage rules in this document, refer to the latest *IPC/JEDEC J-STD-020C*.

7.3 Baking

It is not necessary to bake chips if the conditions specified in section 7.2.2 and section 7.2.3 have not been exceeded. It is necessary to bake chips if any condition specified in section 7.2.2 and section 7.2.3 has been exceeded.

It is necessary to bake chips if the storage humidity condition has been exceeded, we recommend that the device sample removed from its shipment bag for more than 2 days shall be baked to guarantee production.

Baking conditions: 125°C, 8 hours, nitrogen protection. Note that the baking should not exceed 1 times due to a risk of deformation.

8 Reflow Profile

All Allwinner chips provided for clients are lead-free RoHS-compliant products.

The reflow profile recommended in this document is a lead-free reflow profile that is suitable for pure lead-free technology of lead-free solder paste. If customers need to use lead solder paste, contact Allwinner FAE.

The following figure shows the appropriate reflow profile.

Figure 8-1 Lead-free Reflow Profile

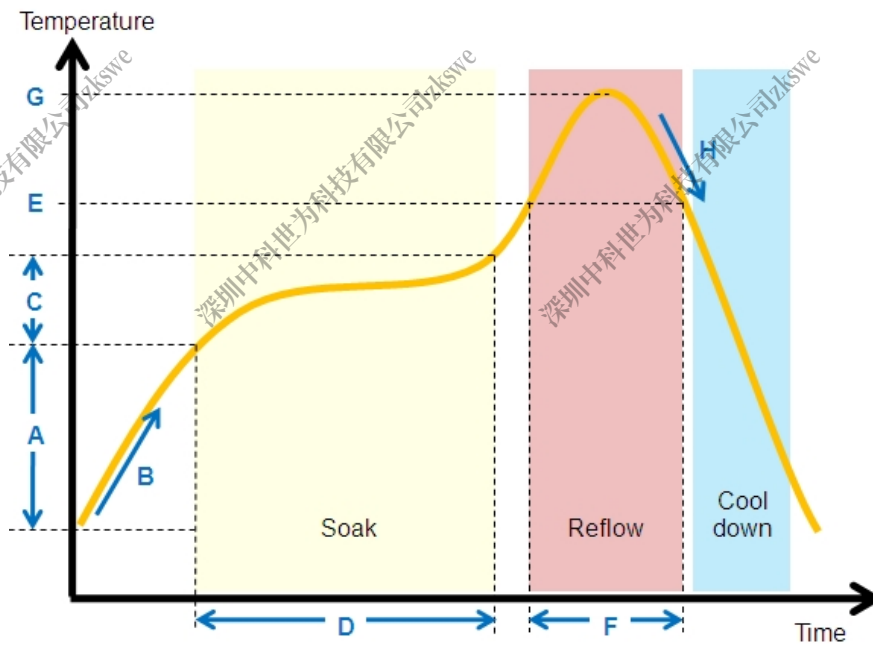


Table 8-1 Lead-free Reflow Profile Conditions

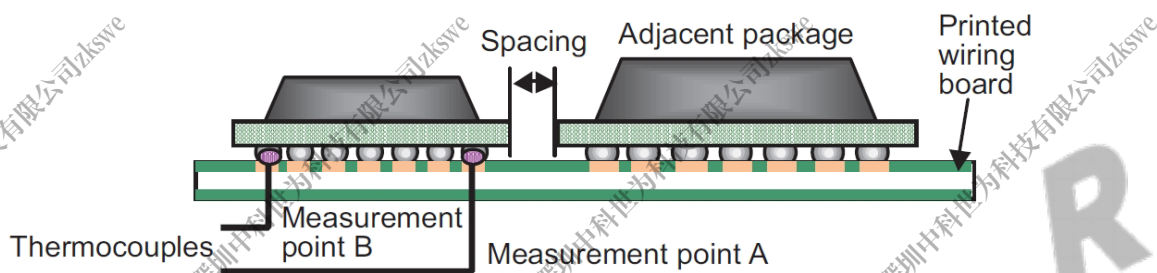
	QTI typical SMT reflow profile conditions (for reference only)	
	Step	Reflow condition
Environment	N2 purge reflow usage (yes/no)	Yes, N2 purge used
	If yes, O2 ppm level	O2 < 1500 ppm
A	Preheat ramp up temperature range	25 °C -> 150 °C
B	Preheat ramp up rate	1.5–2.5 °C/s
C	Soak temperature range	150 °C -> 190 °C
D	Soak time	80–110 s
E	Liquidus temperature	217°C
F	Time above liquidus	60–90 s
G	Peak temperature	240–250 °C

	QTL typical SMT reflow profile conditions (for reference only)	
	Step	Reflow condition
H	Cool down temperature rate	$\leq 4^{\circ}\text{C/s}$

The method of measuring the reflow soldering process is as follows.

Fix the thermocouple probe of the temperature measuring line at the connection point between the pin (solderable end) of the packaged device and the pad by using high-temperature solder wire or high-temperature tape, fix the packaged device at the pad by using high-temperature tape or other methods, and cover over the thermocouple probe. See Figure 10 2.

Figure 8-2 Measuring the Reflow Soldering Process



NOTE

To measure the temperature of the QFP-packaged chip, place the temperature probe directly at the pin.

If possible, the more accurate measuring way is to drill the packaged device, or drill the PCB, and fix the thermocouple probe through the drilled hole at the pad.

9 Part Marking

9.1 F136M2-XXX

The following figure shows the F136M2-XXX marking.

Figure 9-1 F136M2-XXX Marking



The following table describes the F136M2-XXX marking definitions.

Table 9-1 F136M2-XXX Marking Definitions

No.	Marking	Description	Fixed/Dynamic
1	ALLWINNER	Allwinner logo or name	Fixed
2	F136M2-XXX	Product name	Fixed
3	LLLLLL	Lot number	Dynamic
4	XXXXX	Date code	Dynamic

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